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DEPARTMENT OF COMPUTER SCIENCE

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A Hard Negative-Story Threshold

Training-Period Association, Temporal Non-Replication and Economic Limits

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ABSTRACT

Financial-news models often average all stories about a company on a given day, discarding their distribution. This dissertation tests whether negative-story share, the fraction classified as negative, adds information beyond that mean.

I use FinBERT, a language model for financial sentiment, to score news for more than 700,000 US firm-days. A firm-day is one company on one trading day. I compare nine daily summaries during the 2011–2019 training period, then test the selected measure after controlling for mean sentiment, news volume, recent returns and market exposure. The 2020–2023 testing period asks whether the relationship persists. Separate Reuters/LSEG data test whether it appears in another source, while trading costs and risk checks assess practical value.

Negative-story share alone survives correction across the nine training comparisons. A higher share is associated with lower returns after the main controls, but the effect is small and was selected in the same data. It appears during the assigned session and in an earlier window, but not after the close. Without exact story arrival times, the analysis cannot tell whether the news predicted returns, arrived during the return window or was assigned to the wrong session.

Because 48.3% of training firm-days contain one story and estimates on multi-story days are inconclusive, the result does not establish that a mix of stories within a day drives the association.

In the fixed 2020–2023 test, the estimate changes sign. The daily rule cannot cover plausible costs, and the risk rule does not reduce maximum drawdown, the largest fall from a previous peak. The Reuters/LSEG estimates are too uncertain to show that the result appears in another source. The evidence therefore shows a small training-period association with uncertain timing that does not persist or support a practical trading rule.

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CHAPTER 1

INTRODUCTION

1.1 THE AGGREGATION PROBLEM

Several news stories can appear about the same company in one trading session, but there is only one return observation for that firm and session. Comparing news with returns therefore requires one daily news measure. Different summaries retain different information.

The usual summary is the arithmetic mean of the sentiment scores. It measures the average tone, but opposing stories can cancel out. This dissertation instead tests *negative-story share*: the fraction of stories whose strongest FinBERT label is negative. The share measures how much of the day’s coverage is negative. The mean measures the tone of the average story.

Here, “hard” means that each story is assigned to the class with the highest FinBERT probability. It does not mean that the analysis estimates a confidence cut-off.

Consider four hard labels: $(-1, -1, +1, +1)$. Their mean is zero, while their negative-story share is one half. Replacing the two positive labels with neutral labels leaves the share unchanged but makes the mean more negative. Four mildly negative stories, by contrast, produce a share of one even if every continuous score is only slightly below zero. The statistics are related but not interchangeable.

Earlier research links negative financial language to returns, earnings information and slow price adjustment (Tetlock, 2007; Tetlock et al., 2008; Uhl, 2014; Heston and Sinha, 2017).

Research using public news also reports drift after negative coverage and reversal after large price moves with no identifiable news story (Chan, 2003; Savor, 2012). Together, these studies justify comparing daily news summaries. They do not show that negative-story share is useful. The measure must add information beyond average sentiment and story count, precede the return, and survive a later test.

Story count also affects interpretation. The training sample contains 512,153 firm-days, but 48.3% contain one story and the median is two. On a one-story day, negative-story share is simply a binary FinBERT label. It describes a within-day distribution only when several stories are present. Results are therefore reported for all firm-days and for multi-story days.

The statistical and economic questions remain separate. An association can be precisely estimated yet too small to cover trading costs, and a risk rule may reduce losses merely by reducing market exposure. Timing also matters. News published after a return window opens cannot predict that return. The tests therefore ask whether the association exists, when it appears, whether it survives later, and whether it supports a practical rule.

1.2 RESEARCH QUESTIONS AND HYPOTHESES

The dissertation first studies the training period, then asks whether the unchanged relationship survives in a later testing period. The later test uses choices set in advance. FNSPID, a public financial-news and price dataset, is used in both periods. Reuters news supplied through LSEG provides a smaller second sample.

Training period. After allowing for average sentiment and news volume, is a higher negative-story-share rank associated with a lower market-adjusted return rank from the assigned open to the following open? If so, which part of that return window contains the relationship?

Testing period. Does the unchanged baseline relationship remain negative in the fixed 2020–2023 FNSPID news-and-price sample?

Every coefficient compares firms within the same session. Firms are ranked by market-adjusted return, negative-story share, average sentiment and story count. The model asks whether a higher negative-share rank is associated with a lower return rank after controlling for the other news measures. Daily coefficients are averaged so that heavily covered sessions do not dominate. Later models also control for recent returns, volatility and market sensitivity.

A negative coefficient means that firms with more widespread negative coverage tend to rank lower by return. It does not imply that a given increase in negative share causes a fixed price fall. Unobserved news or market conditions may affect both variables, so the result remains a conditional association.

Four hypotheses carry the tests, with the first split into a training-period leg and a testing-period leg.

H1a. In the training period, the negative-share coefficient is below zero after controlling for average sentiment and story count.

- H1b. The unchanged baseline coefficients for all firm-days and multi-story days remain below zero during 2020–2023 after correcting for the two tests.
- H2. A simple high-minus-low negative-share sort within average-sentiment groups produces a negative training-period spread.
- H3. A daily long-short rule earns a positive average return after charging 10 basis points per side on its own turnover.
- H4. After costs, an LSEG risk rule passes its pre-specified return, risk and stability tests against a training-selected control with the same average market exposure.

A high-minus-low sort compares firms with high and low negative-story shares. Turnover is the amount of the portfolio that must be traded. Ten basis points is 0.10% of the amount traded on each side.

H1a is exploratory because the nine-rule training comparison selected negative share. The correction covers the nine initial tests, but the usual interval around the winning estimate does not capture all selection uncertainty.

H1b is narrower. The two coefficients, the multiple-testing adjustment and the decision rule were fixed before the 2020–2023 estimates were calculated. The test is therefore more informative than the training comparison. However, some portfolio outcomes from those years had already been examined, so the period was not completely unseen. Appendix A records the origin and run status of every notebook used in the chapters that follow.

1.3 SUMMARY OF FINDINGS

Only negative-story share remained statistically significant after correcting the nine training-period comparisons. Firms with a higher negative share tended to rank lower by assigned-window open-to-next-open abnormal return. After controlling for average sentiment and story count, the coefficient was -0.00831 , with a 95% confidence interval of $[-0.01410, -0.00252]$ that allows for dependence between nearby sessions.

The timing checks weaken a forecasting interpretation. The relationship appears from the assigned open to close, disappears overnight, and is also present from the previous open to the assigned open. FNSPID lacks reliable row-level availability times, so some stories may have entered a session after its return began. The result is a relationship within the assigned window, not evidence of a forecast based on news known before the return began.

The relationship also fails to persist. In the fixed 2020–2023 testing period, the unchanged coefficient changes sign to $+0.00583$. A direct comparison finds a statistically significant difference between the training and testing estimates. Aggregate data cannot distinguish among changes in timing, coverage, firm composition or classifier outputs.

The economic evidence also argues against practical usefulness. Across the observed negative-share rank spread, the full-control estimate corresponds to -0.512 return-rank percentile points. The best break-even cost among the nine rules is 0.625 basis points per side, versus the assumed 10 basis points.

The separate Reuters/LSEG samples are too small to show that the relationship appears in another source. No risk or prompt rule passes its pre-specified criteria. The training coefficient is therefore a statistical pattern, not a reliable trading signal.

Chapter 2 reviews the literature, Chapters 3 and 4 describe the data and methods, Chapter 5 reports the results, and Chapter 6 concludes.

CHAPTER 2

LITERATURE REVIEW

The statistical literature connects four issues: information in financial news, sentiment measurement, information lost through aggregation and the evidence needed for prediction. A separate literature on risk controls explains why an economic test must compare risk rules with controls that have similar market exposure.

2.1 RETURN INFORMATION IN NEWS TEXT

News stories select facts and compare company results with expectations. A sentiment model therefore measures framing, not fundamentals alone.

Tetlock (2007) finds that pessimism in a daily Wall Street Journal column tracks downward market pressure and high volume before partially reversing. The reversal distinguishes a temporary price move from a lasting change in value. At the firm level, Tetlock et al. (2008) find that the fraction of negative words predicts earnings and returns. Their measure counts words within a document, whereas this dissertation counts negative stories within a firm-day.

The response to a price move depends on its source. Chan (2003) finds drift after identifiable public news, especially bad news, but reversal after large returns without an attached story. Savor (2012) reports a similar split using analyst information. Negative coverage in this study could therefore follow an earlier decline rather than precede a later one. The recent-return controls in Chapter 4 test this reversal explanation.

Coverage and market conditions also change through time. Fang and Peress (2009) link attention to expected returns, while Engelberg and Parsons (2011) show that newspaper distribution affects trading around earnings. García (2013) finds that the predictive content of news language is concentrated in recessions. Audience, coverage and market state are therefore possible sources of temporal change.

Reuters studies motivate the separate LSEG analysis. Using 3.6 million articles, Uhl (2014) finds greater index-level forecasting power in negative than positive sentiment. With

more than 900,000 Reuters stories, Heston and Sinha (2017) report daily predictability, weekly effects and slower responses to negative news. Although their designs differ from this firm-day comparison, both support asking whether an average sentiment score discards useful information.

2.2 CONVERTING TEXT TO A NUMERICAL SCORE

The scoring method determines which language is positive, neutral or negative. General-purpose dictionaries can misclassify financial terms that sound negative in ordinary English but are neutral in company reporting. Loughran and McDonald (2011) discuss terms such as “liability” and “tax”, which a general dictionary can mark as negative even when they are routine in a 10-K. A finance-specific dictionary changes the labels and results. Loughran and McDonald (2016) also identify corpus design, negation, document length and task-level validation as consequential choices.

Contextual models use the words around a phrase rather than counting words in isolation. FinBERT adapts the BERT language model to financial text and classifies sentences as positive, neutral or negative. ProsusAI FinBERT is trained further on Financial PhraseBank, which records labels and disagreement among financially trained readers (Malo et al., 2014; Araci, 2019). A separate finance-trained FinBERT outperforms dictionaries and conventional machine learning in Huang et al. (2023).

This study fixes one FinBERT version throughout. The model was trained on short financial sentences, but it is applied here to FNSPID headlines, Reuters stories and later periods outside that benchmark. Those differences create a measurement limitation.

Prompted large language models can target firm-specific economic effects rather than generic tone. Kirtac and Germano (2024) compare dictionaries, BERT-family models and large language models on nearly a million US financial-news articles. Lopez-Lira and Tang (2026) ask whether a headline is good or bad for the named firm and find delayed responses concentrated in negative news and smaller firms, with weaker results later. These studies motivate the prompt tests, which ask whether a tailored score beats FinBERT on the same dates after costs.

Scoring old news with a modern model can mix look-ahead bias with distraction from company knowledge. Glasserman and Lin (2023) compare original and company-anonymised headlines to separate these effects and find that anonymisation improves in-sample performance. Every prompted result here is therefore a 2024 training-period experiment. Returns never enter the prompt, and the experiment stops if none of the prompts meets the pre-specified criteria.

2.3 THE INFORMATION LOST BY AVERAGING

Each firm-day needs one signal, but document scores can be averaged, selected or modelled separately. The studies reviewed here do not make a like-for-like comparison of simple firm-day summaries. This study compares nine return-blind measures using the same outcome, sample and adjustment for multiple tests.

Equal weighting gives a routine update the same weight as a first release and allows positive or neutral coverage to offset a negative item. A threshold measure keeps that negative item visible.

Earlier work also looks beyond average tone. Allee and DeAngelis (2015) relate the placement of positive and negative words within conference calls to firm performance, reporting choices and user responses. Isakin and Pu (2023) link news-sentiment dispersion to corporate-bond returns and uncertainty, while Chen et al. (2024) study dispersion around Chinese merger announcements. These measures differ from a firm-day share but support retaining distributional information.

Selection is another response to crowded news. Uhl and Novacek (2021) construct a selected “top news” measure and report that investors can benefit from ignoring most news topics. This design instead keeps every story and fixes nine summaries before a shared return test, so return relevance cannot determine which stories enter.

Several stories may describe the same event. Stale and reprinted stories receive a different market response and can be followed by reversal (Tetlock, 2011). Repeated updates about one adverse event also raise negative share without adding information. Story count measures volume, but FNSPID has no stable story-family or publisher identifier. The training regression therefore cannot separate novelty from editorial repetition, which limits both the association and its later non-replication.

2.4 FROM A COEFFICIENT TO AN INVESTMENT CLAIM

Training-period significance does not show that a result will persist. Many proposed return predictors are unstable in later data (Welch and Goyal, 2008). Across 97 cross-sectional predictors, McLean and Pontiff (2016) find smaller estimates outside the original sample and after publication. Sampling error, selection and the tendency to report unusually strong initial results can all contribute.

The order in which data become available also matters. In the exchange between Farmer et al. (2023) and Cakici et al. (2025), replacing a calculation that uses future observations with one that uses only past observations sharply weakens the reported forecasting value. A changed coefficient in this dissertation shows non-replication, but cannot identify whether the market, sample or measurement changed.

An information coefficient is the daily rank correlation between a signal and returns. Information coefficients and portfolio returns answer different questions: a precise association may still be too small to trade. Korajczyk and Sadka (2004) show that proportional costs and price impact materially change momentum profitability and capacity, while Novy-Marx and Velikov (2016) find the worst damage among high-turnover anomalies. The economic tests report gross and net return, turnover and per-side break-even cost.

When several predictors are tested, some may look significant by chance (Harvey et al., 2016). The Benjamini–Hochberg (BH) adjustment limits the expected share of false findings within a declared group of tests, under its stated assumptions (Benjamini and Hochberg, 1995). It covers the nine-rule comparison and related groups, not specifications designed after seeing results. The training coefficient is therefore treated as exploratory.

Nearby sessions can be related, and their variance can change. Newey–West heteroskedasticity and autocorrelation consistent (HAC) standard errors allow for both features (Newey and West, 1987). Portfolio comparisons use fixed-length block resampling to keep nearby sessions together (Künsch, 1989); this implementation wraps the blocks at the sample boundary. It differs from the random-length stationary bootstrap of Politis and Romano (1994).

An interval crossing zero can reflect either a small effect or low precision. Minimum detectable effects (MDEs) quantify the effect size a test has a good chance of detecting (Bloom, 1995). If a short LSEG sample can detect only coefficients several times larger than the FNSPID estimate, its null cannot establish whether an FNSPID-sized relationship exists in LSEG.

2.5 EVALUATING RISK OVERLAYS

A weak directional signal might still identify periods for lower risk. However, time in cash mechanically lowers volatility and losses, so the control must have similar market exposure.

The sentiment-free benchmark is a heterogeneous autoregressive (HAR) volatility forecast. Corsi (2009) models realised volatility with daily, weekly and monthly components. In the benchmark used here, exposure falls as the forecast rises. Evidence on inverse-volatility factor scaling (Moreira and Muir, 2017) motivates this benchmark without endorsing sentiment timing.

The first comparator is the fixed HAR path. The second holds a constant fraction of that path to match the news rule’s average exposure. In FNSPID, the constant is calibrated after observing the 2020–2023 testing results, so it supports only descriptive lower-exposure attribution. In LSEG, matched constants are selected in the training folds before forward testing.

A circular-shift check moves the full risk-off schedule through returns while preserving its activation frequency. It tests whether the same-session risk schedule aligns unusually

with downside outcomes, but not whether news was observable before returns began. A timing claim needs both unusual alignment and verified pre-return availability. The matched constant separately tests whether lower average exposure explains the result.

2.6 THE RESEARCH GAP

The measurement comparison uses one outcome, a declared test family and dependence-aware uncertainty. An investment claim also needs testing-period evidence and trading costs. This dissertation applies those requirements to a hard-threshold firm-day measure through training selection, timing, testing, economic and separate-source checks. The analysis retains null results without using them to start a further search.

CHAPTER 3

DATA AND PANEL CONSTRUCTION

3.1 THE FNSPID PANEL

FNSPID combines about 15.7 million public financial-news records with 29.7 million US equity-price observations (Dong et al., 2024). I retain firms with coverage in every calendar year.

The fixed rule requires at least 20 mappable news days in every calendar year from 2011 through 2023. It returns 574 ticker strings. Three lack price files and one produces no joined firm-day, leaving 570 after joining news, adjusted opens and SPY prices. SPY is an exchange-traded fund that tracks the S&P 500 and is used here as the market benchmark. This coverage-filtered firm universe favours surviving, well-covered firms. It follows data availability rather than historical index membership and applies no further security-type filter.

The panel contains 715,546 news-bearing firm-days across 3,262 sessions: 512,153 in training and 203,393 in testing. Four training outcomes are missing, so 512,149 rows enter the nine-rule comparison.

Training-period coverage is sparse: 247,470 firm-days (48.3%) have one story, 127,541 (24.9%) have two, and 137,142 (26.8%) have at least three. On a one-story day, negative share is a binary label rather than a distribution across stories.

Each story row stores its symbol, assigned session, whether it is marked as a recap and three FinBERT class probabilities. Rows are collapsed into the firm-session counts, means, dispersion, label shares and other rules in Chapter 4. Each observation combines one priced firm with one assigned exchange session; this is the analytic unit (i, t) . Story text remains in a local database and never enters this repository or its intermediate files.

Timing follows the fixed session-mapping rule. Date-only records move to the first New York Stock Exchange session after their date, whereas timestamped records keep the session implied by that time. The date-only rule can therefore delay pre-open news by a full session. Because the data contain neither verified availability times nor a row-level

timing-precision flag, it is not possible to identify the affected rows. Some stories may have appeared after their assigned return window opened, which prevents an interpretation based on news known before the return began.

3.2 SCORING A STORY

FinBERT assigns each story a probability of positive, neutral and negative sentiment. The same fixed ProsusAI FinBERT classifier scores both data sources. They use model identifier `ProsusAI/finbert` at revision `4556d13015211d73dccd3fdd39d39232506f3e43`, loaded from an enforced local snapshot. For story j about firm i on assigned session t , the continuous score is

$$s_{ijt} = p_{ijt}(\text{positive}) - p_{ijt}(\text{negative}), \quad -1 \leq s_{ijt} \leq 1. \quad (3.1)$$

The hard label is the class with the highest probability. A negative probability of 0.40 wins if the other two are lower, so the boundary is relative rather than a confidence threshold. The model was pretrained on financial text and fine-tuned on Financial PhraseBank (Araci, 2019). FNSPID has no human-label audit or independent-model comparison, so the analysis concerns the unverified outputs of one fixed classifier.

The mix of sentiment classes changes modestly at the period boundary. The 1,183,961 training stories are 16.80% negative, 60.01% neutral and 23.20% positive. The 415,490 testing stories are 15.84%, 61.42% and 22.74%. Negative share is zero on 74.93% of training firm-days and 76.85% of testing firm-days. Single-story days rise from 48.32% to 56.53%. These shifts do not explain the sign reversal alone. Timing, mapping, coverage, classifier behaviour and composition may also matter.

Training covers 2,264 sessions from 3 January 2011 to 31 December 2019. Testing covers 998 sessions from 2 January 2020 to 18 December 2023. Before the testing estimates were calculated, the two coefficients, multiple-testing adjustment, power assumptions and decision rule were fixed. The test then ran once without changes.

Some later portfolio outcomes had already been examined. This decision was recorded internally rather than in an independently timestamped preregistration, so the testing period was not completely unseen.

3.3 THE RETURN BEING EXPLAINED

Daily prices come from the FNSPID archive, whose reference collection code downloads Yahoo Finance bars. Yahoo publishes an adjusted close but not an adjusted open, so I build the proxy¹

$$P_{it}^{\text{adj,open}} = P_{it}^{\text{open}} \frac{P_{it}^{\text{adj,close}}}{P_{it}^{\text{close}}}. \quad (3.2)$$

Applying Yahoo’s adjusted-close ratio to the open transfers its split and dividend adjustments. I did not reconstruct corporate actions independently. The result is therefore neither an audited total-return series nor an exact implementation price. Prices advance to the next SPY-calendar session, not the firm’s next news day, avoiding multi-session returns when an intervening session has no story. The primary outcome is

$$r_{i,t+1}^A = \left(\frac{P_{i,t+1}^{\text{adj,open}}}{P_{it}^{\text{adj,open}}} - 1 \right) - \left(\frac{P_{\text{SPY},t+1}^{\text{adj,open}}}{P_{\text{SPY},t}^{\text{adj,open}}} - 1 \right). \quad (3.3)$$

Subtracting SPY’s open-to-open return gives a simple market-adjusted event-study outcome (MacKinlay, 1997). Two additional sensitivity checks replace unit market exposure with a 252-session trailing stock-SPY beta. One beta-adjusts the outcome; the other also controls for beta rank. The baseline association and period contrast persist, but remain rank associations, not estimates of factor alpha.

3.4 THE REUTERS/LSEG DATA

Reuters coverage supplied through LSEG covers 33 large US firms across several sectors. It is denser and has richer source and availability fields than the available FNSPID data, making it useful as a separate second sample.

The two non-overlapping LSEG periods both follow the FNSPID training period. The earlier period has 14,935 firm-days across 456 sessions (2 January 2024–24 October 2025). The later period has 5,511 rows across 167 sessions (27 October 2025–26 June 2026). Both cover 33 firms.

News density prevents a direct replication. Median daily story count is 61 in the earlier period and 90 in the later period, versus 2 in FNSPID training. A large share change may flip one of two FNSPID labels, but shifts a dense distribution in LSEG.

LSEG maps each story to the first New York Stock Exchange (XNYS) open at least fifteen minutes after recorded availability, which FNSPID cannot do. The cross-source table uses raw open-to-open returns because the LSEG aggregate lacks the FNSPID abnormal-

¹See the FNSPID price scraper and Yahoo adjusted-close documentation, both accessed 15 August 2026.

return construction. A common centred percentile-rank transform puts the coefficients on a comparable scale. The outcomes remain different and are labelled accordingly. The sources are never pooled.

The firm-level LSEG economic test requires all 33 firms to be present in every included session. This removes 11 sessions from the earlier period. The complete panel contains 612 sessions: 445 for training and 167 for testing. The training sessions are divided into three chronological folds without changing their time order. A single rule selected from these folds is then tested on the later period.

The rule is compared with an equal-weight long portfolio and two constant-exposure controls selected in training.

Licensed source records, complete request payloads and raw model responses are excluded from this repository. The prompt appendix reproduces the available templates and describes the headline fields sent to the model provider.

3.5 DATA LIMITATIONS

FNSPID lacks stable publisher identifiers, near-duplicate clusters and links from an update to its first release. Several negative rows may therefore restate one event. Controlling for $\log(1 + n_{it})$ separates score patterns from coverage volume, but not new information from repetition, a distinction that affects market response (Tetlock, 2011).

Source density differs sharply, classifier errors remain possible, and FNSPID requires news, a mapped symbol, a stock open and a SPY open. Average firms per session fall from 226.2 in training to 203.8 in testing. There is no common-universe re-estimate, publisher breakdown or row-level timestamp audit. The period difference shows that the estimated relationship is not stable across these periods, but it does not explain why. Market conditions, coverage, production, mapping and classifier behaviour all remain possible explanations.

CHAPTER 4

METHODOLOGY

4.1 THE ANALYSIS IN SIX STEPS

The analysis has six ordered steps. A measure reaches the economic tests only after the fixed statistical comparison, preventing a later return result from replacing the measure chosen at the start.

1. Compare nine firm-day summaries using one outcome and one multiplicity correction.
2. Test whether the selected measure adds to average sentiment.
3. Split the return window into parts to see where the relationship appears.
4. Apply the unchanged baseline once to the fixed 2020–2023 testing period.
5. Translate the coefficient into return scale, portfolio performance, turnover and costs.
6. Evaluate a same-session FNSPID risk diagnostic, then repeat the coefficient and economic work in the LSEG data.

No statistic covers association, timing and practical value. Rank correlations measure co-movement. Portfolio returns require turnover and costs. A drawdown claim also requires matched market exposure and verified signal availability.

The selected coefficient is exploratory because the aggregation analysis changed before the final research question was set. Comparing its p -value with $0.05/9 = 0.00556$ provides a conservative nine-test check, but does not capture all uncertainty from choosing the best rule (Harvey et al., 2016). Some later checks were also designed after seeing the association.

The testing-period estimates, test groups and decision rule were fixed before they were calculated. However, some outcomes from those years had already been examined.

4.2 STEP 1: NINE FIRM-DAY AGGREGATION RULES

Let $\mathcal{J}_{it}$ contain the stories assigned to firm i on session t , with $n_{it} = |\mathcal{J}_{it}|$. The continuous score s_{ijt} comes from Equation (3.1), and $h_{ijt} \in \{-1, 0, 1\}$ is the hard negative, neutral or positive label. Eight rules use only the current firm-day. The ninth carries a decayed state through time.

1. **Mean hard label:** $a_{it}^{\text{hard}} = n_{it}^{-1} \sum_j h_{ijt}$.
2. **Mean continuous score:** $a_{it}^{\text{mean}} = n_{it}^{-1} \sum_j s_{ijt}$.
3. **Median continuous score:** $a_{it}^{\text{median}} = \text{median}_j(s_{ijt})$.
4. **Ten-per-cent trimmed mean:** sort the scores, drop $k = \lfloor 0.1n_{it} \rfloor$ observations from each tail and average the rest. With small n_{it} , $k = 0$ and the rule collapses to the mean.
5. **Negative-story share:**

$$q_{it}^- = \frac{1}{n_{it}} \sum_{j \in \mathcal{J}_{it}} \mathbf{1}(h_{ijt} = -1). \quad (4.1)$$

6. **Dispersion:** the sample standard deviation of s_{ijt} , set to zero on a one-story firm-day.
7. **Strongest event:** prefer a non-recap story, then take the remaining story with the largest $|s_{ijt}|$. An exact tie in opposite directions is recorded as missing, and the return never breaks a tie.
8. **Log-count attention:** $a_{it}^{\text{attn}} = a_{it}^{\text{mean}} \log(1 + n_{it})$. This rule picks out no particular story. It gives more cross-sectional weight to a mean built from more stories.
9. **Decayed state:** convert the current continuous mean into an impulse $u_{it} = \text{sign}(a_{it}^{\text{mean}})|a_{it}^{\text{mean}}|^{1.5}$. Writing d_{it-} for the state at firm i 's previous news-bearing session and Δ_{it} for the number of exchange sessions since it,

$$d_{it} = 2^{-\Delta_{it}/3} d_{it-} \left(1 - \frac{3}{4} \mathbf{1}[u_{it} d_{it-} < 0] \right) + u_{it}. \quad (4.2)$$

Tone persists across sessions, and a strictly opposite non-zero impulse resets three quarters of the decayed prior state before the new impulse enters. The state is initialised at zero for each firm's first news-bearing session.

The rules are deliberately simple: learning an aggregator would add another selection problem and obscure what the mean discarded. The candidates cover a threshold, dispersion, trimmed extremes, story selection, attention and a cross-session state.

For each session, firms are ranked by the signal and by return. The Spearman correlation asks whether firms with higher signal ranks also tend to have higher or lower return ranks. For rule a , it is

$$IC_t^a = \text{corr}_S(a_{it}, r_{i,t+1}^A). \quad (4.3)$$

The reported coefficient is $\overline{IC}^a = T^{-1} \sum_t IC_t^a$, requiring at least three complete firms per session. Five-lag Newey-West HAC errors allow changing variance and short-run dependence (Newey and West, 1987). The nine two-sided p -values form one Benjamini-Hochberg family at $q = 0.05$ (Benjamini and Hochberg, 1995). Story-count groups are descriptive. Expected directions are fixed: positive-score summaries point upward, while negative share and dispersion point downward.

4.3 STEP 2: THE RELATIONSHIP AFTER CONTROLLING FOR THE MEAN

For each session, firms are ranked by return, negative share, average sentiment and story count. The regression asks whether negative-share rank is still related to return rank after accounting for the other two news measures. The formal calculation uses within-session percentile ranks, with tied values given their average rank and each rank shifted by one half. Let N_t be the available firms, L the number strictly below a firm on x , and E its tie-group size:

$$\tilde{x}_{it} = R_t(x_{it}) - \frac{1}{2}, \quad R_t(x_{it}) = \frac{L + \frac{1}{2}(E + 1)}{N_t}. \quad (4.4)$$

Without ties, shifted ranks span $N_t^{-1} - \frac{1}{2}$ to $\frac{1}{2}$. Their mean is $1/(2N_t)$, which the daily intercept absorbs. Because ties narrow the support, effect translations use observed rank spreads. The base model is

$$\tilde{r}_{i,t+1}^A = \alpha_t + \beta_t \tilde{q}_{it}^- + \gamma_t \tilde{s}_{it} + \delta_t \widetilde{\log(1 + n_{it})} + \varepsilon_{it}, \quad (4.5)$$

where $\bar{s}_{it}$ is average continuous sentiment and tildes denote shifted ranks. The reported coefficient averages the daily slopes:

$$\bar{\beta} = \frac{1}{T} \sum_{t=1}^T \beta_t. \quad (4.6)$$

A session needs ten complete firms and an estimable design matrix. The slope series uses a five-lag HAC error. Ranking limits outliers and gives FNSPID and LSEG a common scale. Since ties prevent a full unit change in negative-share rank, translations use the observed spread.

The price-path specification adds three shifted ranks to test the reversal finding of Chan (2003):

$$\tilde{r}_{i,t+1}^A = \alpha_t + \beta_t \tilde{q}_{it} + \gamma_t \tilde{s}_{it} + \delta_t \log(\widetilde{1 + n_{it}}) + \theta_{1t} \tilde{r}_{it}^{(1)} + \theta_{5t} \tilde{r}_{it}^{(5)} + \phi_t \tilde{\sigma}_{it}^{(20)} + \varepsilon_{it}. \quad (4.7)$$

Here $r^{(1)}$ is the adjusted-open return from $t - 1$ to t , $r^{(5)}$ is the cumulative return from $t - 5$ to t , and $\sigma^{(20)}$ is the standard deviation of the twenty returns ending at formation. Every control ends before the outcome starts. The full model is the main reversal test. Intermediate rows show the effect of adding controls.

Three post-result measurement checks form a separate Benjamini–Hochberg family. The first uses a close-to-close market-adjusted outcome. The second further removes the component explained by a trailing Fama–French three-factor (FF3) model (Fama and French, 1993). The third keeps the primary outcome but replaces hard negative share with average $p(\text{negative})$. Factor betas use the 252 sessions ending at formation, require 126 observations and exclude the outcome session.

The rank control is linear. On one-story days, negative share is a thresholded class from the same probability vector as mean sentiment, so its coefficient may capture a nonlinear boundary rather than aggregation. The result is conditional only on the stated controls. A more flexible model would be another training-sample test and is left for a prospectively specified study.

On price-complete rows, the pooled 90th-minus-10th percentile negative-share-rank spread is 0.56056. The mean session spread is 0.52722. Multiplying either by $100\bar{\beta}$ gives return-rank percentile points. A hypothetical change of 0.8 exceeds the observed pooled range and is not used.

4.4 STEP 3: LOCATING THE ASSOCIATION INSIDE THE RETURN WINDOW

The timing test splits stock and SPY returns into assigned-open-to-close, assigned-close-to-next-open and previous-close-to-assigned-open parts, plus a separate previous-open-to-assigned-open window. Gross-return identities are checked to 10^{-12} , and the three new training legs form one correction family. The testing-period results and contrasts between periods are additional checks rather than pre-specified tests. Without story arrival times, the split cannot establish sequence.

A second check uses a 252-session rolling stock-SPY beta ending at the assigned open and requiring 126 observations. One model beta-adjusts the outcome. Another also controls for beta rank. The two tests are corrected within each period. A separate

correction is applied to the two between-period contrasts. Neither analysis estimates causality or tradable alpha.

4.5 STEP 4: THE FIXED 2020–2023 TEST

The fixed testing procedure evaluates two pre-specified families once. The first repeats the nine two-sided information-coefficient tests from Equation (4.3). The second repeats Equation (4.5) on all firm-days and the pre-specified multi-story subset, with a shared correction and negative expected direction. It does not examine testing-period portfolios, alternative outcomes, price controls, thresholds, event types, scorers or other story-count groups.

Before the two fixed 2020–2023 coefficient families were calculated, an amendment added prospective power information and a direct training-versus-testing contrast. Expected precision scales the training HAC error by $\sqrt{2,264/998}$ and uses the corrected two-test threshold, a two-sided size of 0.025. A descriptive check uses the observed testing error instead.

To compare the periods, I stack the unchanged daily coefficient series and estimate

$$\beta_t = \alpha + \Delta D_t^{\text{test}} + u_t \quad (4.8)$$

with HAC(5) uncertainty. D_t^{test} is zero in training and one in testing, so $\alpha = \bar{\beta}_{\text{training}}$ and $\Delta = \bar{\beta}_{\text{testing}} - \bar{\beta}_{\text{training}}$. The contrast uses Equation (4.5), excluding the price-path model. It is needed because significance in one period and non-significance in another do not prove a difference. It does not change the verdict from the primary tests.

4.6 STEP 5: TRANSLATION INTO ECONOMIC TERMS

This step covers the double-sort comparison, portfolio costs and learned no-trade filters.

4.6.1 DOUBLE-SORT COMPARISON

The double sort first groups firms with similar average sentiment. Within each group, it then compares firms with low, middle and high negative-story shares. The high-minus-low difference shows whether negative share matters among firms with similar average sentiment. Fractional ties prevent symbol order or returns from breaking them. Cells need five firms and sessions need three usable groups. The daily high-minus-low series uses HAC(5) uncertainty and is repeated for multi-story days. Its p -value is descriptive. Earlier fixed cut-offs produced no usable sessions, so the tie-aware replacement was fixed before estimation.

4.6.2 RANK PORTFOLIOS, TURNOVER AND COSTS

Each rule also forms a daily cross-sectional rank portfolio. Expected-direction ranks are scaled to equal long and short gross exposure. Positions formed at open t earn abnormal returns to open $t + 1$. Let w_{it} be the target weight, r_{it} the raw stock return into open t , and w_{it}^{pre} the previous target after that return:

$$w_{it}^{\text{pre}} = \frac{w_{i,t-1}(1 + r_{it})}{1 + \sum_j w_{j,t-1}r_{jt}}. \quad (4.9)$$

Turnover is reported one-way, as half the traded notional:

$$\text{TO}_t = \frac{1}{2} \sum_i |w_{it} - w_{it}^{\text{pre}}|. \quad (4.10)$$

A quoted per-side cost applies to both sales and purchases, giving a charge of $2c \text{TO}_t$ and a net return of

$$r_{p,t+1}^{\text{net}} = \sum_i w_{it} r_{i,t+1}^A - 2c \text{TO}_t. \quad (4.11)$$

Starting a dollar-neutral portfolio with gross exposure of one gives $\text{TO}_t = 0.5$ and costs c . Final liquidation charges the same rate on half the remaining absolute notional. The main cost is 10 basis points per side. Descriptive break-even cost is $\bar{r}_p^{\text{gross}}/(2\overline{\text{TO}})$, annualised turnover is $252\overline{\text{TO}}$, and Sharpe ratios use the same annualisation. A non-positive denominator in Equation (4.9) implies insolvency and leaves later cost metrics undefined unless returns are capped.

4.6.3 LEARNED NO-TRADE FILTERS

Learned no-trade thresholds test sensitivity to the simple rule. Logistic regression, gradient boosting and a multilayer perceptron (MLP) predict whether to trade. The comparison covers testing-period discrimination, measured by the area under the receiver operating characteristic curve (AUC), trading activity and net return against a fixed band, an activity-matched rule and cash. A network fitted to shuffled labels is the negative control.

4.7 STEP 6: SAME-SESSION RISK TESTS

The risk analysis asks whether weak directional information coincides with periods when lower exposure would have helped. The FNSPID exercise is a same-session diagnostic, not a strategy test based on information known in advance, because the data do not show when each story became available.

4.7.1 VOLATILITY BENCHMARK

The benchmark reduces market exposure when recent volatility points to higher risk. The news rule reduces exposure further on unusually negative-news days. It adds value only if it improves downside outcomes beyond the volatility rule at a similar level of market exposure.

The sentiment-free HAR benchmark forecasts the average squared SPY return over the next five sessions. At decision open t , $u_t = r_{\text{SPY},t-1 \rightarrow t}^2$ is the previous session's squared adjusted-open return and is already known. The forecast target is $v_t^{(5)} = 5^{-1} \sum_{j=0}^4 r_{\text{SPY},t+j \rightarrow t+j+1}^2$. With $\epsilon = 10^{-8}$, training-period ordinary least squares (OLS) estimates

$$\log(v_t^{(5)} + \epsilon) = \alpha + \beta_1 \log(u_t + \epsilon) + \beta_5 \log\left(\frac{1}{5} \sum_{j=0}^4 u_{t-j} + \epsilon\right) + \beta_{22} \log\left(\frac{1}{22} \sum_{j=0}^{21} u_{t-j} + \epsilon\right) + e_t. \quad (4.12)$$

Let x_t collect the three logged volatility measures in Equation (4.12), excluding the intercept. Converting the forecast back from logs gives $\hat{v}_t = \kappa \exp(\hat{\alpha} + x_t' \hat{\beta})$, where the positive training-period scale correction is $\kappa = \text{mean}\{v_t^{(5)} / \exp(\hat{\alpha} + x_t' \hat{\beta})\}$; the mean is taken over training-period targets. This correction adjusts the average forecast level after the conversion from logs.

The fixed test model is fitted only to five-session targets whose windows end by 31 December 2019. A historical sensitivity starts after 504 rows and refits expanding OLS every 21 sessions using completed target windows. Forecasts are converted into exposure as $\text{clip}\{0.10/\sqrt{252\hat{v}_t}, 0.25, 1\}$, so higher forecast volatility reduces market exposure. Exposure cannot fall below 0.25 or exceed 1. Cash earns zero, costs are 2 basis points per side, and the overlay cannot exceed HAR exposure.

4.7.2 SAME-SESSION NEWS RULE

The sentiment input is market-wide negative-news pressure: the headline-count-weighted negative share, compared with its recent history.

$$P_t = \frac{\sum_i q_{it}^- n_{it}}{\sum_i n_{it}}, \quad z_t = \frac{P_t - \mu_{t-1}}{\sigma_{t-1}}, \quad (4.13)$$

Here μ_{t-1} and σ_{t-1} use the 252 sessions ending at $t-1$ and require 126 observations. Current-session news enters P_t and the numerator of z_t . Only the normalisation is historical. The signal is therefore not lagged. A two-threshold state defines the exposure modifier:

$$S_t = \begin{cases} 1 & \text{if } S_{t-1} = 0 \text{ and } z_t \geq 1.5, \\ 0 & \text{if } S_{t-1} = 1 \text{ and } z_t \leq 0.5, \\ S_{t-1} & \text{otherwise,} \end{cases} \quad (4.14)$$

with $S_0 = 0$ and the prior state retained when z_t is unavailable. When $S_t = 1$, HAR exposure is multiplied by 0.25. Separate entry and exit thresholds limit switching. The state calculated for session t sets exposure in that same session; it is not delayed by one session. The result therefore cannot be interpreted as a strategy based on information known before the session began.

4.7.3 MATCHED CONTROLS AND UNCERTAINTY

For a like-for-like comparison, a constant multiplier rescales HAR to the news rule’s mean 2020–2023 exposure. This scale is calculated from the testing results, so it provides an after-the-fact explanation of lower exposure rather than a control selected in training.

Paired block bootstraps keep nearby sessions in fixed twenty-session blocks and wrap them at the sample boundary, using 4,999 or 9,999 repetitions as recorded in each manifest (Künsch, 1989). Return differences and reductions in squared negative returns are separate outcomes. Circular shifts test unusual alignment of the same-session risk schedule with downside outcomes, but cannot establish signal availability. Additional checks omit each episode and February–April 2020 in turn.

4.8 THE LSEG ANALYSIS AND THE PROMPT EXPERIMENTS

The unchanged FNSPID pressure rule is first applied to both LSEG periods. A restricted search then tests other aggregate rules under a fixed plan. The selected aggregate rule is compared with HAR and a constant multiplier chosen in training.

A separate firm-level rule is compared with full investment and with fixed company weights that match its training exposure. A rule passes only if it improves tail risk or utility, retains an acceptable lower bound on return and is stable across checks. Half-sample and leave-one-firm-out results show whether the finding depends on a small part of the data.

Prompt experiments compare generic tone with firm-specific economic labels. Five Gemma 4 26B prompts target one-session reaction, delayed reaction, cash-flow revision, surprise catalysts and a red-team consensus view. A sixth targets the first tradable price reaction. Each returns a score on a fixed grid. Reuters headlines and company names passed through OpenRouter to DeepInfra under recorded no-retention controls. Returns were excluded from every payload.

Training uses 2024. A prompt is retained only if it meets five pre-specified conditions. It must have a positive net mean after 10 basis points per side, a positive net Sharpe and

a break-even cost of at least 10 basis points. It must also be positive in three quarters and beat the same-date Gemma baseline in three quarters. If none meets all five conditions, the procedure computes no 2025 prompt portfolio. Appendix B records the tested wording because the null depends on what the model was asked.

4.9 REPORTING NULL RESULTS

A null result may mean that there is little effect, or that the sample is too small to detect it. The minimum detectable effect (MDE) is the smallest effect that the test has an 80% chance of detecting under the stated assumptions. For standard error SE , the nominal two-sided 5% MDE is

$$MDE_{0.80} = (z_{0.975} + z_{0.80}) SE = 2.8016SE. \quad (4.15)$$

For m tests, the conservative first-BH threshold replaces $z_{0.975}$ with $z_{1-0.05/(2m)}$. This is sufficient for the first ordered test, not exact adaptive-BH power. MDEs are reported for LSEG transfer, story-type, earnings-window, publisher and learned-threshold nulls. The analysis records families specified before the main analysis and labels later families separately.

CHAPTER 5

RESULTS

Results follow the six methodology steps. Threshold and story-count checks come after the temporal test because they affect both periods.

5.1 STEPS 1 AND 2: AGGREGATION COMPARISON AND CONTROLLED ESTIMATE

Only negative-story share remains statistically significant after correcting the nine training-period comparisons, and the relationship is small and negative. It is the only rule in Table 5.1 and Figure 5.1 to survive the adjustment for nine comparisons: its mean daily rank correlation is -0.00543 , with HAC $t = -3.12$ and $p = 0.0018$. The hard-label and continuous-score means are near 0.003 , both with $p > 0.09$. Median sentiment and decayed state have only uncorrected p -values below 0.05 . Non-mean summaries do not outperform as a group. The comparison also cannot explain why negative-story share alone survived correction.

The relationship remains negative after accounting for average sentiment and story count. Across 512,149 firm-days and 2,264 sessions, the controlled negative-share coefficient is -0.00831 (SE 0.00295 , 95% CI $[-0.01410, -0.00252]$, $p = 0.0049$). It clears the simple nine-test Bonferroni threshold of 0.00556 .

The same training data helped select negative share before the model was estimated. This interval therefore understates selection uncertainty, and the estimate may be inflated.

The coefficient describes an ordering, not a basis-point return. Within a session, higher negative-share rank is associated with lower assigned-window return rank after controlling for mean sentiment and $\log(1 + n)$. Adding one- and five-session return ranks plus 20-session volatility gives -0.00914 (SE 0.00282 , 95% CI $[-0.01466, -0.00362]$, $p = 0.00118$) on price-complete rows. Table 5.2 shows the nested models.

Table 5.1. Nine aggregation rules in the FNSPID training period. $\overline{IC}$ is the mean daily Spearman rank correlation with the assigned-session open-to-next-open abnormal return, and t uses Newey–West HAC(5). BH applies Benjamini–Hochberg correction across the nine rules. Break-even is the per-side cost that makes the corresponding rank portfolio’s mean net return zero. The rules use 512,145–512,149 firm-days across 2,264 sessions.

Aggregation rule	$\overline{IC}$	HAC t	p	BH	Break-even (bps)
Negative-story share	−0.00543	−3.12	0.0018	✓	0.625
Decayed sentiment state	0.00416	2.23	0.0259	–	0.323
Median continuous score	0.00347	2.02	0.0431	–	0.320
Log-count weighted mean	0.00328	1.81	0.0704	–	0.245
Trimmed mean	0.00302	1.70	0.0889	–	0.248
Mean continuous score	0.00298	1.68	0.0930	–	0.245
Mean hard label	0.00280	1.58	0.1149	–	0.292
Strongest event	0.00271	1.55	0.1205	–	0.116
Score dispersion	−0.00149	−0.92	0.3590	–	0.278

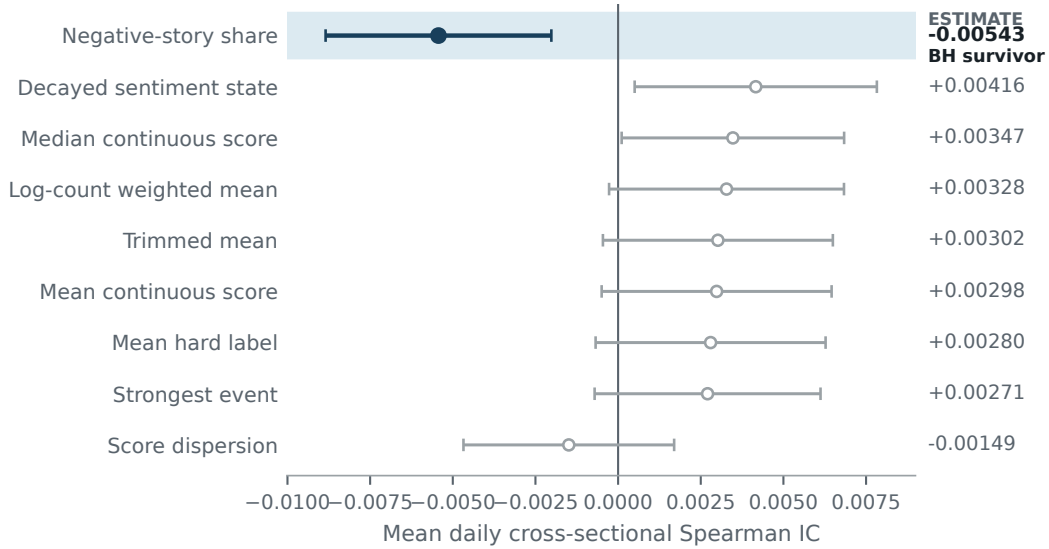


Figure 5.1. Mean daily rank correlation between each aggregation rule and the assigned-session open-to-next-open abnormal return in the FNSPID training period. Bars are 95% Newey–West HAC(5) intervals over 2,264 sessions. Grey rules do not remain significant after correction across nine tests. Source: aggregate Notebook 03 outputs.

Table 5.2. Negative-story-share coefficient as recent-return and volatility controls are added. All variables use average-tie percentile ranks shifted by one half within session.

Specification	$\hat{\beta}_{neg}$	HAC SE	95% CI	p
Full-sample baseline	−0.00831	0.00295	[−0.01410, −0.00252]	0.0049
Price-complete baseline	−0.00830	0.00296	[−0.01410, −0.00251]	0.0050
+ one-session return	−0.00945	0.00291	[−0.01515, −0.00375]	0.0012
+ one- and five-session returns	−0.00978	0.00288	[−0.01541, −0.00414]	0.0007
+ returns and 20-session volatility	−0.00914	0.00282	[−0.01466, −0.00362]	0.0012

In the full model, the one-session return and volatility controls are both negative, while the mean-sentiment coefficient is close to zero (Table 5.2). Negative share remains negative after these controls are added. The beta-adjusted coefficient is -0.01054 , or -0.00950 after controlling for beta rank; both survive correction.

Ties restrict the observed negative-share-rank spread to 0.56056 between the pooled 90th and 10th percentiles. The full-control translation is therefore $0.56056 \times -0.00914 = -0.00512$, or -0.512 return-rank percentile points. Using the mean within-session spread gives -0.482 points. Both are small.

Table 5.3 collects the main training-period, testing-period, scale and separate-source coefficients.

Table 5.3. Main negative-story-share estimates. FNSPID uses assigned-session open-to-next-open abnormal returns; LSEG uses raw open-to-open returns under its recorded availability rule. The two FNSPID testing-period conditional rows are corrected together. The difference between periods is a separate check fixed in advance. Where shown, intervals are 95% HAC intervals.

Test	Estimate	95% interval	p or q	Sessions
Aggregation-family IC	-0.00543	–	$p = 0.0018$	2,264
Conditional baseline	-0.00831	$[-0.01410, -0.00252]$	$p = 0.0049$	2,264
Full price-path controls	-0.00914	$[-0.01466, -0.00362]$	$p = 0.0012$	2,264
Testing period: all firm-days	0.00583	$[-0.00426, 0.01593]$	$q = 0.515$	998
Testing period: at least two stories	0.00295	$[-0.00990, 0.01581]$	$q = 0.653$	998
Testing minus training	0.01414	$[0.00250, 0.02579]$	$p = 0.0173$	3,262
Model-free high-low (bps)	-0.618	$[-1.510, 0.275]$	$p = 0.175$	2,264
LSEG earlier-period coefficient	-0.02861	$[-0.07283, 0.01560]$	$q = 0.409$	456
LSEG later-period coefficient	-0.01573	$[-0.09453, 0.06308]$	$q = 0.696$	167

5.2 STEP 3: WHERE THE ASSOCIATION APPEARS IN THE RETURN WINDOW

The window split in Table 5.4 changes the timing interpretation. The assigned-open-to-close coefficient is -0.01029 (95% CI $[-0.01596, -0.00462]$, BH $q = 0.0011$), while close-to-next-open is $+0.00155$ ($[-0.00416, +0.00725]$, $q = 0.595$). The training association is intraday, not in the later overnight leg that more plausibly follows the day’s news.

Pre-open returns do not resolve the problem. Previous-close-to-assigned-open is -0.00638 (unadjusted $p = 0.0457$, BH $q = 0.0686$). The separate previous-open-to-assigned-open probe is -0.01926 (95% CI $[-0.02580, -0.01272]$, $p = 7.9 \times 10^{-9}$). Stock and SPY return identities reconcile within 1.8×10^{-15} , and the reconstructed abnormal return matches the baseline panel exactly.

Table 5.4. Post-hoc FNSPID return-window decomposition. The assigned-open-to-next-open row repeats the baseline and is excluded from the new multiplicity adjustment. Within each period, BH correction covers the three decomposed return legs. The three between-period differences form a separate family. The previous-open analysis is a separate post-hoc timing diagnostic. All intervals use HAC(5).

Outcome or contrast	Estimate	95% CI	p	BH q
<i>Training period</i>				
Assigned open to next open (reference)	−0.00831	[−0.01410, −0.00252]	0.0049	–
Assigned open to close	−0.01029	[−0.01596, −0.00462]	0.0004	0.0011
Assigned close to next open	0.00155	[−0.00416, 0.00725]	0.5949	0.5949
Previous close to assigned open	−0.00638	[−0.01264, −0.00012]	0.0457	0.0686
Previous open to assigned open (separate)	−0.01926	[−0.02580, −0.01272]	7.9×10^{-9}	–
<i>Testing period (additional timing diagnostic)</i>				
Assigned open to next open (reference)	0.00583	[−0.00426, 0.01593]	0.2574	–
Assigned open to close	0.00337	[−0.00669, 0.01343]	0.5111	0.5111
Assigned close to next open	0.00348	[−0.00616, 0.01312]	0.4796	0.5111
Previous close to assigned open	−0.01552	[−0.02521, −0.00582]	0.0017	0.0051
<i>Testing minus training</i>				
Assigned open to next open (reference)	0.01414	[0.00250, 0.02579]	0.0173	–
Assigned open to close	0.01366	[0.00211, 0.02521]	0.0204	0.0613
Assigned close to next open	0.00193	[−0.00928, 0.01314]	0.7357	0.7357
Previous close to assigned open	−0.00914	[−0.02067, 0.00240]	0.1205	0.1807

The FNSPID data cannot place each story before or after the assigned open, so they do not support a claim that the effect is specific to next-open returns. The intraday coefficient may be contemporaneous. Assignment lag or earlier information may explain the previous-open result. Aggregate data cannot separate these explanations or quantify their role in the story-count and period patterns.

The post-result measurement checks in Table 5.5 carry no timing claim. The hard-share coefficient is +0.00027 (BH $q = 0.929$) on close-to-close returns and +0.00182 ($q = 0.815$) when the outcome is the trailing FF3 residual of those returns. Mean FinBERT $p(\text{negative})$ is −0.00378 on the primary outcome ($q = 0.337$). None survives correction.

Table 5.5. Measurement checks added after the main FNSPID training-period result. The baseline row repeats the earlier next-open hard-share estimate and is not a fourth test. BH is Benjamini–Hochberg at $q = 0.05$ across the three new coefficients. Close-to-close starts at the assigned session close; FF3 residuals use trailing 252-session betas that exclude the outcome session.

Test	Estimate	95% CI	p	q	BH
Assigned-window hard share (baseline)	−0.00831	[−0.01410, −0.00252]	0.0049	–	–
Close-to-close hard share	0.00027	[−0.00572, 0.00627]	0.9292	0.929	–
FF3 residual hard share	0.00182	[−0.00404, 0.00767]	0.5435	0.815	–
Assigned-window soft negative mass	−0.00378	[−0.00844, 0.00088]	0.1122	0.337	–

5.3 STEP 4: THE FIXED 2020–2023 TEST

The fixed test does not repeat the training-period relationship: the unchanged coefficient changes sign. Across 203,393 testing firm-days and 998 sessions, the all-firm-day coefficient

is $+0.00583$ (95% CI $[-0.00426, +0.01593]$, $p = 0.257$, BH $q = 0.515$). The pre-specified multi-story estimate is $+0.00295$ ($[-0.00990, +0.01581]$, $q = 0.653$). None of the nine information coefficients survives correction, so H1b is not supported. Under the pre-specified decision rule, no alternative aggregator is selected and the signal is not reversed.

The pre-specified testing-minus-training contrast is $+0.01414$ (SE 0.00594 , 95% CI $[+0.00250, +0.02579]$, $p = 0.0173$). It remains statistically significant with HAC lags from 0 to 42. The daily coefficient series shows almost no serial dependence, and Ljung–Box tests through lag 21 find no evidence of serial correlation. The contrast is therefore unlikely to be caused by the chosen lag length.

Changing the market benchmark does not remove the contrast. It is $+0.01884$ after trailing-beta adjustment and $+0.01628$ with beta rank controlled; both survive correction. Unit market beta does not explain the period difference.

The window split does not locate the change. Testing-period intraday and post-close coefficients are small and positive, while the pre-open overnight coefficient is negative and survives correction. None of the corresponding between-period contrasts survives correction. These additional tests do not identify one return leg as the source of the instability.

Figure 5.2 places both periods on one axis. Annual means are negative from 2011 to 2018 and non-negative from 2019 onward. Because 2019 is the final training year, the sign change does not align with the fixed boundary and starts before COVID-19. Every annual interval crosses zero, so the figure identifies neither a break date nor its cause.

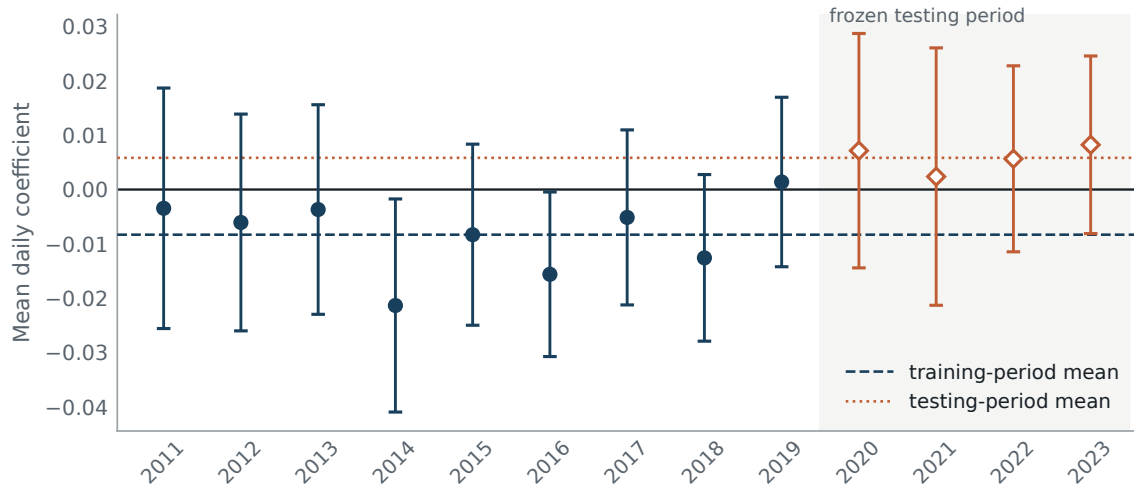


Figure 5.2. Calendar-year mean of the daily negative-share coefficient from Equation (4.5). Bars are 95% intervals from annual standard errors. The daily coefficients have almost no autocorrelation, so these are close to HAC intervals. Dashed and dotted lines mark the training and testing means. The figure is descriptive: annual estimates are uncertain and show neither a break date nor a cause. Source: aggregate Notebook 75 daily coefficient series.

The test had limited ability to detect a small negative relationship. Pre-test power was 35.4%, with an 80%-power MDE of 0.01372 at the corrected two-test threshold. The observed HAC error was 15.8% larger, implying 26.5% power and an MDE of 0.01588. The failed test therefore cannot establish a zero effect. Winner selection and sampling variation may contribute to the discrepancy, but the data do not identify the cause. The positive sign also cautions against transferring García (2013)’s aggregate recession result directly to this firm-level design.

5.4 THRESHOLDS AND STORY COUNTS

The baseline remains negative at HAC lags 5, 10 and 21, but story-count subsets are less stable (Table 5.6 and Figure 5.3). The coefficient is -0.01025 on one-story days (95% CI $[-0.01943, -0.00107]$, $p = 0.0286$), -0.00529 at $n \geq 2$ ($[-0.01272, +0.00214]$), and -0.00183 at $n \geq 3$ ($[-0.01385, +0.01019]$).

Table 5.6. How the baseline negative-share coefficient changes with same-day story count (upper panel) and Newey–West lag length (lower panel). All rows use average-tie percentile ranks shifted by one half within session and the assigned-session open-to-next-open abnormal return. Row counts are shown before missing values are dropped; four full-panel rows lack the return.

Variation	$\hat{\beta}_{neg}$	95% CI	p	Panel rows
<i>Same-day story count</i>				
All firm-days (baseline)	-0.00831	$[-0.01410, -0.00252]$	0.0049	512,153
At least two stories ($n_{it} \geq 2$)	-0.00529	$[-0.01272, 0.00214]$	0.1629	264,683
At least three stories ($n_{it} \geq 3$)	-0.00183	$[-0.01385, 0.01019]$	0.7656	137,142
<i>Newey–West lag length</i>				
5 lags (baseline)	-0.00831	$[-0.01410, -0.00252]$	0.0049	512,153
10 lags	-0.00831	$[-0.01414, -0.00248]$	0.0052	512,153
21 lags	-0.00831	$[-0.01407, -0.00255]$	0.0047	512,153

Separate subset models do not test whether groups differ: ranks are recalculated within each subset and the one-story model drops the constant count regressor. Observed-scale effects are -0.516 , -0.324 and -0.122 percentile points, with overlapping intervals. A one-story mechanism remains unresolved. The testing period has more singletons, so that shift alone cannot explain its positive coefficient.

The double sort in Figure 5.4 is similar. Within mean-sentiment quintiles, high-minus-low negative share averages -0.618 basis points per session (95% CI $[-1.510, +0.275]$, descriptive $p = 0.175$). At $n \geq 2$, the estimate is -0.253 ($[-1.821, +1.315]$). H2 has the declared direction, but both intervals include zero.

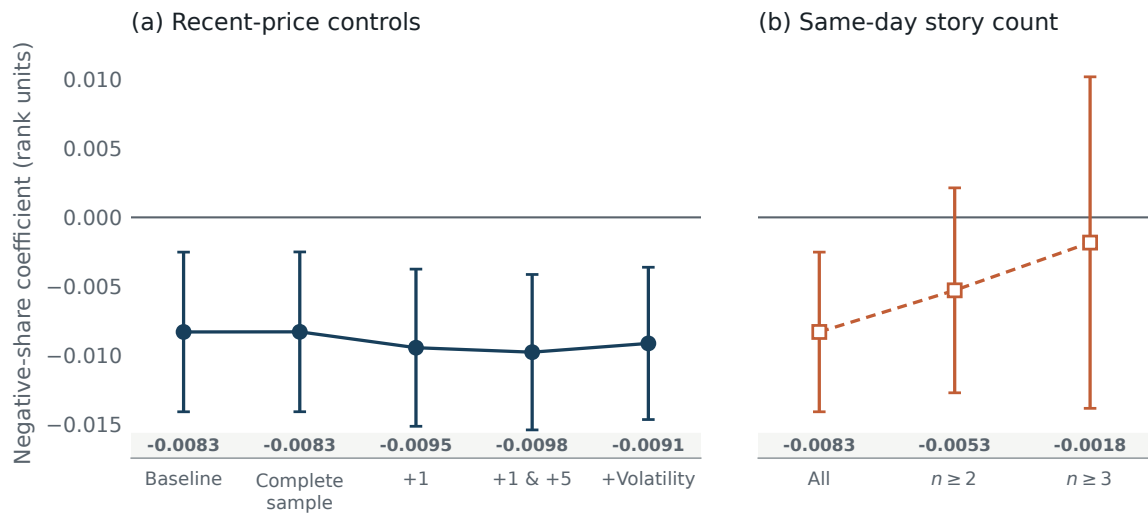


Figure 5.3. The negative-share coefficient as recent-price controls are added (left) and the minimum same-day story count rises (right). Bars are 95% HAC(5) intervals. Sources: aggregate Notebooks 74 and 79.

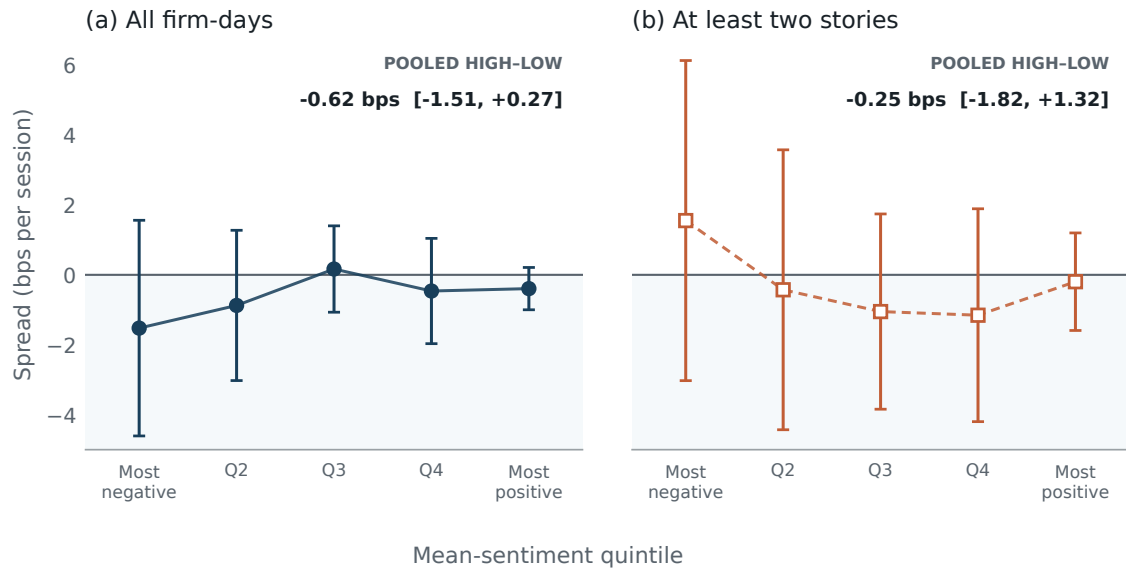


Figure 5.4. High-minus-low difference in assigned-session open-to-next-open abnormal return, calculated within five average-sentiment groups. The right panel requires at least two stories. All 95% HAC(5) intervals cross zero. Source: aggregate Notebook 76 outputs.

5.5 STEP 5: TRANSACTION COSTS

Before costs, the negative-share portfolio earns 0.890 basis points per session on one-way turnover of 0.712 (Figure 5.5). A 10-basis-point per-side charge costs 14.24 basis points and leaves -13.35 basis points net. The descriptive break-even point is 0.625 basis points per side. Across nine rules it ranges from 0.116 to 0.625, making the assumed charge 16.0 times the best point estimate.

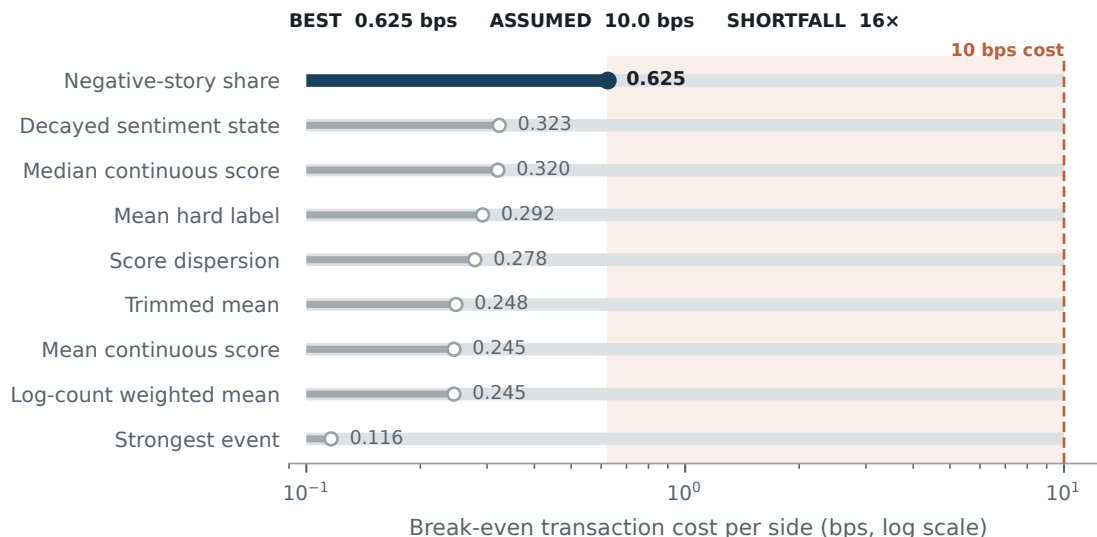


Figure 5.5. Per-side break-even transaction-cost point estimates for nine FNSPID rules on a logarithmic scale. The shaded band is the gap between the best estimate and the 10-basis-point charge. Source: aggregate Notebook 03 outputs.

Learned no-trade filters do not solve the cost problem. Testing AUCs are 0.5005–0.5038, and net performance is 7.72–16.89 basis points below the fixed band. Gradient-boosted and logistic losses exceed their nominal MDEs. The MLP’s 7.72-basis-point loss remains imprecise against a 10.04-basis-point MDE. The shuffled-label control’s simulated wealth falls below zero, so later performance measures are undefined. H3 fails for every implementation.

5.6 STEP 6: SAME-SESSION RISK TESTS

The news modifier does not show a clear overall advantage over the volatility benchmark once their different levels of market exposure are considered.

In the 2020–2023 same-session comparison, the negative-pressure state raises HAR cumulative net return from 29.85% to 35.20% and lowers mean exposure from 0.619 to 0.569. The 0.366-basis-point average return advantage is imprecise, while squared downside falls ($p = 0.0148$). Maximum drawdown worsens from -15.73% to -16.36% , despite improving in 2020 from -12.95% to -7.39% .

An after-the-fact constant HAR multiplier matches the overlay’s realised mean exposure. The overlay gains 0.592 basis points per session against this descriptive comparator (95% block-bootstrap CI $[-0.635, +1.962]$, $p = 0.3712$). The interval crosses zero, and the comparator was not selected in training.

Almost the entire advantage in Figure 5.6 arises during the first ten weeks of 2020. The overlay cuts exposure during the COVID-19 crash and avoids part of the HAR portfolio’s fall to a net-equity index of 0.90. The paths then run almost parallel, and the overlay gives back part of that lead in 2022.

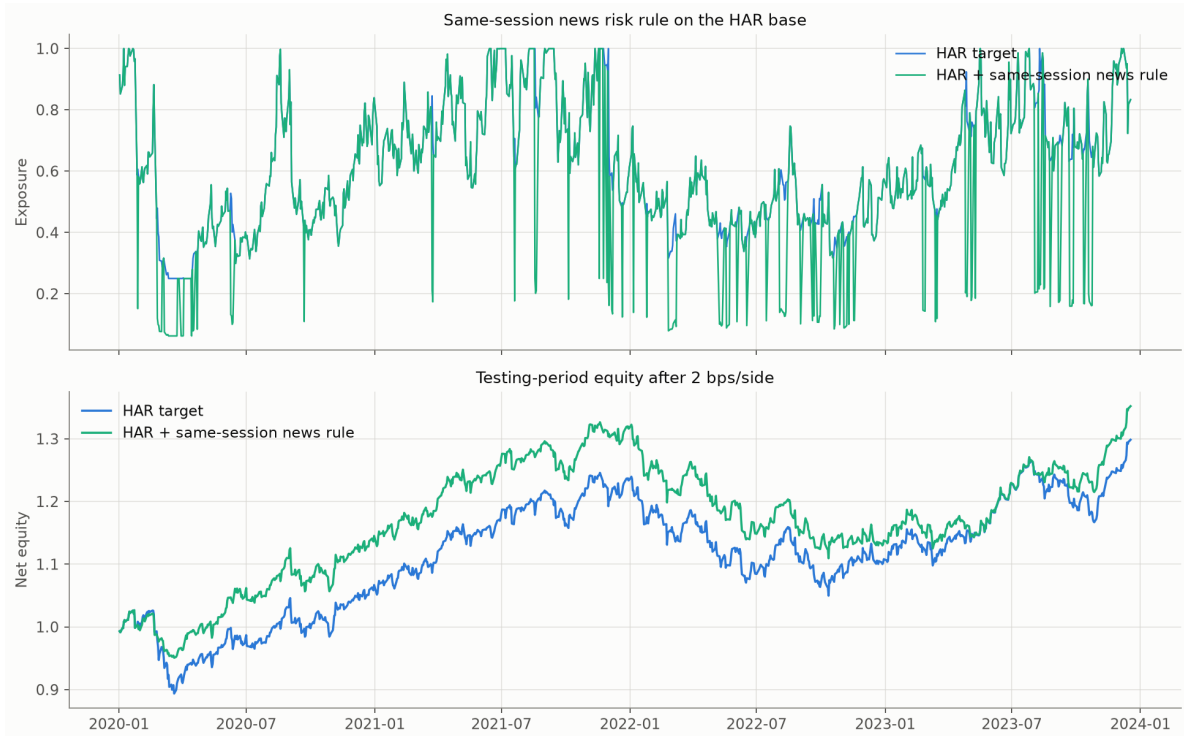


Figure 5.6. Daily exposure (top) and cumulative net equity after 2 basis points per side (bottom) for the fixed HAR target and the same-session news risk rule across the 998-session testing period. The news rule ends at 1.352 against the base at 1.298. It uses news assigned to the current session and is not a one-session-lagged trading signal. Source: Notebook 24.

Figures 5.7 and 5.8 first test whether the risk-off schedule aligns unusually with downside days. They do not show whether the news was known before returns began. About 1% of non-zero circular shifts match the observed FNSPID 2020–2023 squared-downside reduction. Across the three tested periods (FNSPID 2013–2019, FNSPID 2020–2023 and LSEG/Gemma 2025–2026), only FNSPID 2020–2023 remains unusual after correction (BH $q = 0.0301$).

A separate concentration check omits each of the 44 FNSPID 2020–2023 risk-off episodes in turn. Every resulting confidence interval for the squared-downside reduction remains above zero. The interval also remains above zero after excluding February–April 2020

($p = 0.0034$). Although the two largest episodes contribute 30.1% and 14.3% of the total reduction, neither one episode nor the first COVID-19 crash explains the whole result.

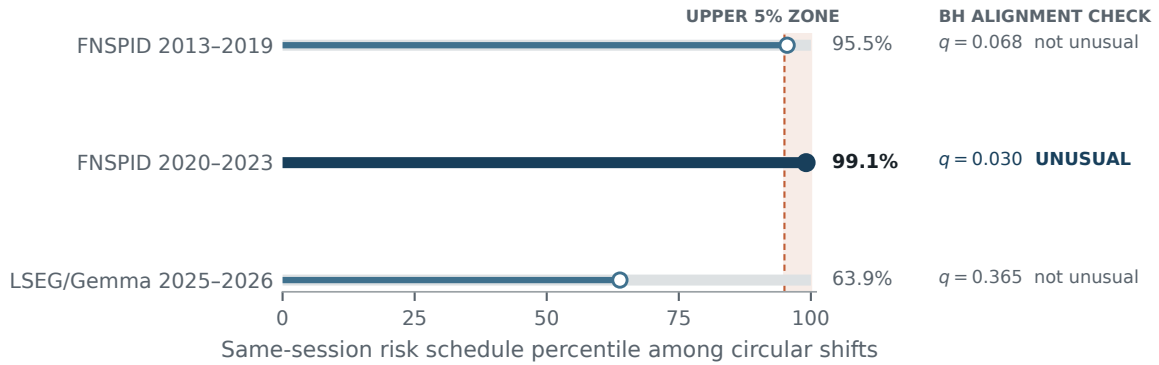


Figure 5.7. Position of each observed squared-downside reduction among non-zero shifts of the same-session risk-off schedule. The three periods are FNSPID 2013–2019, FNSPID 2020–2023 and LSEG/Gemma 2025–2026. Only FNSPID 2020–2023 remains unusual after BH correction. This diagnostic tests alignment conditional on the session mapping; it does not establish that the signal was available before the return. Source: aggregate Notebook 45 outputs.

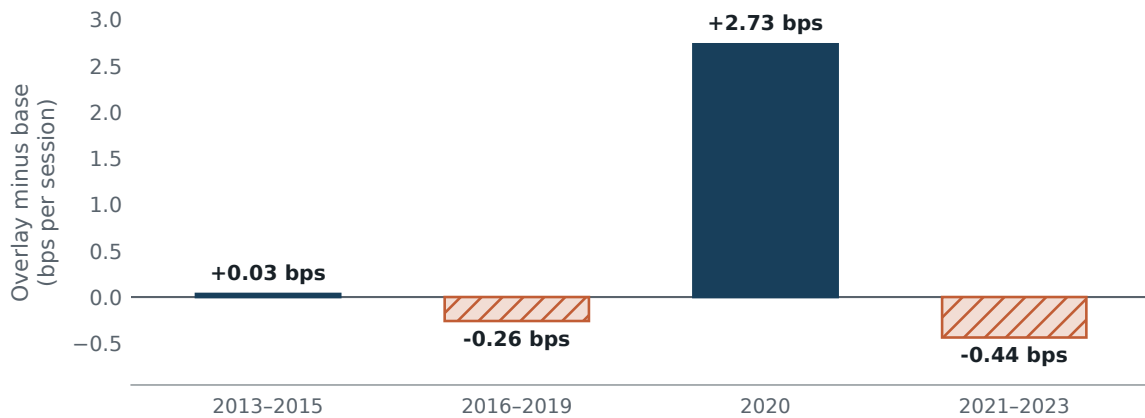


Figure 5.8. Mean return of the same-session news risk rule minus its sentiment-free base by period. These are period means. Only the pooled 2020–2023 squared-downside alignment result survives its correction. Sources: aggregate Notebooks 24 and 25.

The downside result survives schedule-shift and leave-out checks, but it remains a same-session relationship because current-session news sets exposure under the news risk rule. Overall drawdown is worse, return advantages over HAR and the after-the-fact constant are imprecise, the 2013–2019 comparison is null, and the rule never activates in the later LSEG period. It has no demonstrated timing or economic advantage based on information known in advance.

5.7 THE LSEG ANALYSIS

Both primary LSEG coefficients are negative, but they use a different data source and return definition from FNSPID. The earlier-period estimate is -0.02861 (95% CI $[-0.07283, +0.01560]$, BH $q = 0.409$). The later-period estimate is -0.01573 ($[-0.09453, +0.06308]$, $q = 0.696$). Their 80%-power MDEs are 7.6 and 13.6 times the FNSPID baseline. The samples cannot show whether the relationship repeats in another source (Figure 5.9).

An earlier 44-firm LSEG comparison also produces no BH survivor in either of its two families: the 18-test aggregator/scorer family and the eight-test scorer/filter family. It is reported separately because it uses different firms.

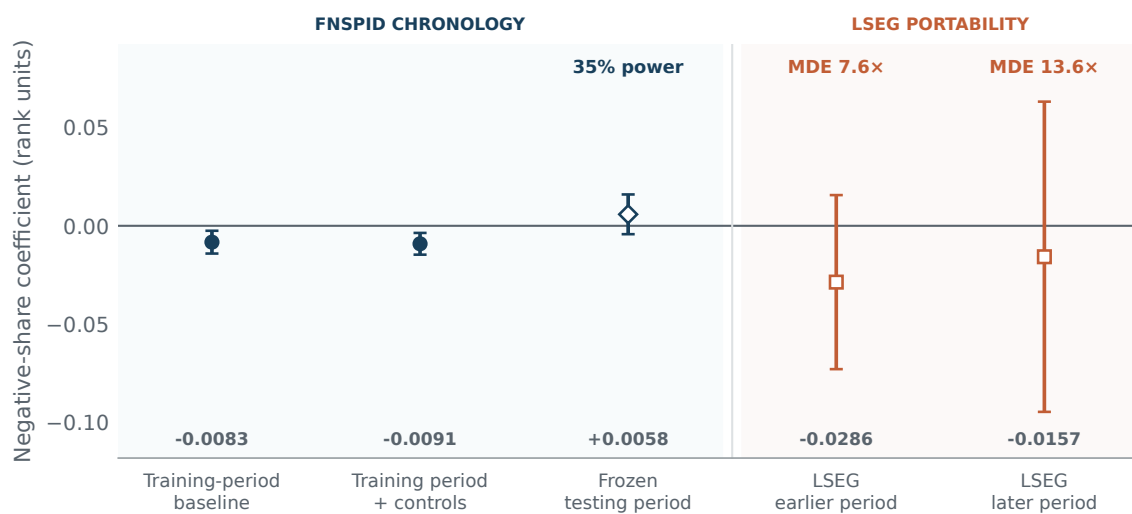


Figure 5.9. Negative-share estimates after the main controls, with 95% HAC intervals. The fixed FNSPID test uses 2020–2023. LSEG uses raw returns rather than FNSPID market-adjusted returns, so the data sources are never combined. LSEG labels show its 80%-power MDE as a multiple of the FNSPID coefficient. Sources: aggregate Notebooks 71, 75, 77 and 79.

The FNSPID risk rule never fires in the later period: its maximum z -score is 1.349 against an entry threshold of 1.5. It fires on 53 of 456 sessions in the earlier period. Downside and shift placement survive correction, but the return difference is 0.144 basis points (95% CI $[-1.295, +1.688]$), and two of eighteen episodes provide 65.6% of the reduction. This does not show that the result repeats in another source.

The aggregate search compared 6,672 combinations of news-pressure measure, historical normalisation, lookback window, risk-off mapping, threshold and exposure multiplier. All comparisons used three chronological 152-session training folds. One aggregate rule was selected from those folds and evaluated once, unchanged, in the later 167-session testing period. Its testing difference is -0.049 basis points (95% CI $[-0.656, +0.633]$).

A separate firm-level search compared 4,608 company-level rules using the 445 complete training sessions described in Chapter 3. The selected rule was then evaluated in the same 167-session testing period. It reduces losses against full investment but loses to training-selected constant company weights, with 0.63 percentage points more drawdown, 0.27 percentage points more downside deviation and 25.9 times the turnover (Figure 5.10).

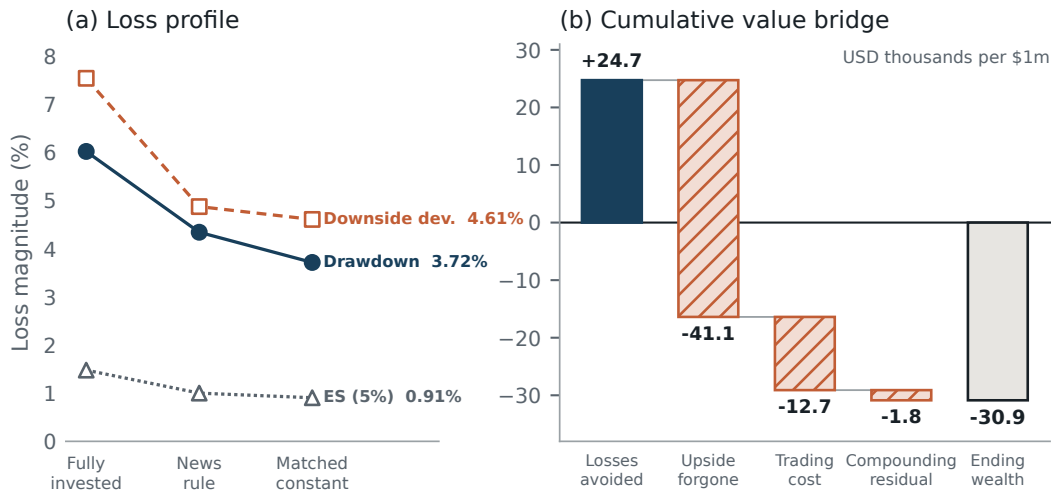


Figure 5.10. Firm-level LSEG loss control over the 167-session testing period. The news rule reduces losses compared with full investment, but a constant rule with the same average exposure performs better. The right panel shows how avoided losses, missed gains, costs and compounding produce the difference in ending wealth. Source: aggregate Notebook 82 outputs.

Per US\$1 million, the rule avoids US\$24,738 of losses but forgoes US\$41,126 of gains. It also incurs US\$12,715 of costs. A US\$1,767 compounding residual reconciles the US\$30,870 ending-wealth gap. Every pre-specified economic criterion fails, so H4 fails.

5.8 PROMPT EXPERIMENTS AND WHAT THE NULLS SHOW

No prompt meets the pre-specified 2024 training criteria (Figure 5.11). The best loses 0.989 basis points per session gross and 4.012 net. The structured reaction score loses 2.590 gross and 19.666 net. All break-even costs are negative, so no 2025 prompt portfolio is computed. Appendix B gives one complete request, the five variant-specific task paragraphs and all per-prompt results.

Many secondary tests are too imprecise to rule out effects that could still matter economically. Median nominal MDEs in Table 5.7 are 18.8 basis points per unit of negative share for story-type interactions, 15.5 for earnings-window interactions and 0.049 daily-IC

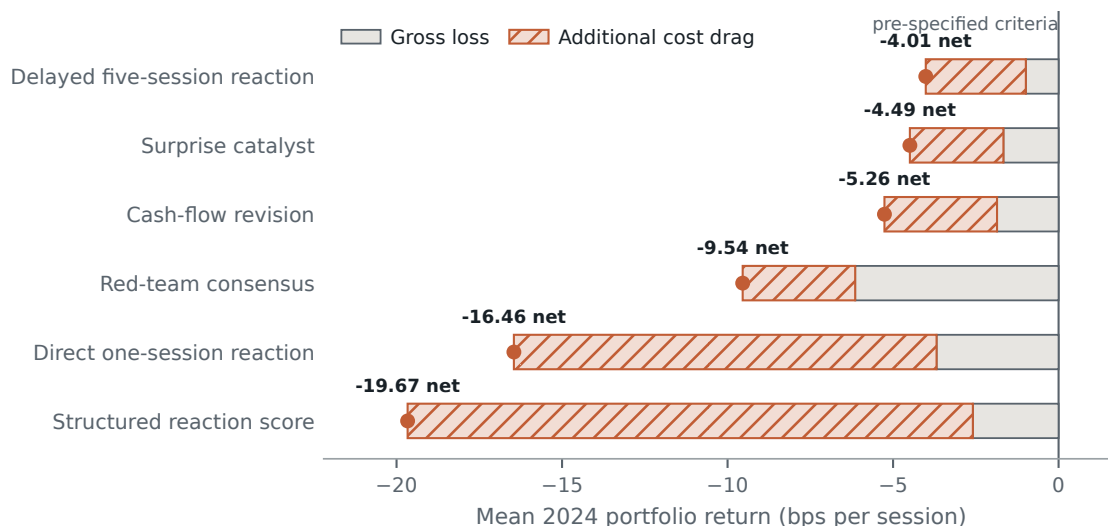


Figure 5.11. Gross and net 2024 portfolio means for five target-specific prompts and one structured reaction score. Every variant is negative before costs and fails the pre-specified criteria. Sources: aggregate Notebooks 83 and 84.

units for publisher rules. These large thresholds leave economically relevant effects inside the null intervals.

Table 5.7. Minimum detectable effects for the principal null families. The conservative column uses the critical z -value implied by the first Benjamini–Hochberg threshold, $z_{1-0.05/(2m)}$, for a family of size m .

Family	Tests	Median MDE	Conservative BH MDE	Unit
Earnings-window conditioning	3	15.519	17.923	bps interaction vs outside
LSEG conditional transfer	2	0.088	0.097	rank coefficient
Learned trade/no-trade thresholds	3	6.883	7.949	net bps/session vs historical fixed band
Publisher conditioning	4	0.049	0.059	daily cross-sectional IC
Story-type conditioning	12	18.802	24.878	bps per unit negative share

Taken together, H1a is consistent with the training estimate, subject to selection and timing. H1b is not supported in testing, H2 has the declared sign but is imprecise, and H3–H4 miss their economic criteria. The evidence establishes neither causality, pre-return availability nor cost coverage.

CHAPTER 6

CONCLUSIONS AND FURTHER WORK

6.1 ANSWERS TO THE RESEARCH QUESTIONS

The training sample contains a negative association after the main controls. Firm-days with a higher negative-story-share rank tend to have a lower assigned-window return rank after allowing for mean sentiment and story count. The baseline coefficient is -0.00831 and remains negative after controlling for recent returns, volatility and recent market sensitivity.

This result provides only exploratory support for H1a. Negative-story share was chosen from nine candidates using the same training data. The adjusted test reduces the risk of a chance finding, but confirmation requires a new sample.

FNSPID does not record reliable story-availability times. The association appears in both the assigned return window and an earlier window. The evidence therefore shows a relationship within mapped return windows, not one based on news known before the return began.

The fixed testing-period result does not confirm the training association. During 2020–2023, the unchanged coefficient changes sign to $+0.00583$. The multi-story estimate is also inconclusive, and no aggregation rule survives correction. The testing-minus-training contrast persists under longer HAC lags and a trailing-beta benchmark. Annual estimates are imprecise and do not identify a break at the chosen boundary. H1b is therefore not supported.

Low power limits what that failure proves. The testing interval still contains small negative effects, so it cannot establish that the true coefficient is zero. It does establish that the pre-specified test did not produce the negative evidence needed to support H1b.

The economic evidence is less ambiguous. The observed-scale effect is small. The high-minus-low return spread has the sign specified by H2, but its interval crosses zero. The best break-even cost is 0.625 basis points per side, versus the assumed 10 basis points. H3 fails: the training association does not fund the turnover needed to trade it.

The risk tests provide no economic evidence in support of the trading rule. In FNSPID, the news rule lowers one downside measure but worsens maximum drawdown. Because current-session news sets current-session exposure, this is a same-session check rather than a strategy formed from information known in advance. Its return advantage over an after-the-fact constant with the same average exposure is also imprecise.

The LSEG rules fail against controls selected in the training folds with the same average exposure, and no prompt variant passes its pre-specified criteria. The samples are too small to show whether the relationship repeats across sources. H4 therefore fails.

Table 6.1 summarises these conclusions.

Table 6.1. Final hypothesis and cross-source verdicts. “Consistent” means the selected training-period estimate has the direction stated in H1a, but it is exploratory rather than a confirmation.

Claim	Verdict	Reason
H1a	Consistent, not confirmed	Negative selected training-period coefficient; timing is intraday and assignment remains ambiguous.
H1b	Not supported	Neither testing-period conditional coefficient is negative, and no testing-period rule survives correction.
H2	Direction only	High-minus-low spread is negative, descriptive and imprecise.
H3	Fails	Turnover makes the daily implementation negative after the declared cost.
H4	Fails	No LSEG risk rule meets all matched-control economic criteria.
Cross-source result	Unresolved	Both LSEG FinBERT intervals cross zero and their MDEs are many times the FNSPID coefficient.

6.2 CONTRIBUTION

The academic contribution is a direct comparison of nine firm-day summaries, followed by a fixed sequence of stronger checks for the selected measure. The checks add controls, locate the association within the return window, repeat it in a fixed testing period and test its economic value. Each one answers a question that the training coefficient cannot answer alone.

Earlier studies link negative financial news with weaker returns (Tetlock et al., 2008; Uhl, 2014; Heston and Sinha, 2017). Other work shows that features beyond an average can matter, including word placement, sentiment dispersion and news selection (Allee and DeAngelis, 2015; Isakin and Pu, 2023; Uhl and Novacek, 2021). The training result agrees with this earlier evidence, but supports only a small, sample-specific association.

The coefficient weakens and becomes inconclusive on multi-story days. FNSPID also cannot distinguish separate events from repeated coverage, even though repeated stories can receive a different market response (Tetlock, 2011). The evidence therefore does not show that a hard threshold is generally better than averaging.

The practical contribution is a clear screening process. A credible news signal must meet three conditions. It must be available before the return begins, persist in later data and remain useful after costs relative to a control with similar market exposure. This signal fails all three conditions. Its timing is uncertain, its sign changes in testing and its return does not cover trading costs.

This result is consistent with wider evidence that return predictors often weaken outside their original samples (Welch and Goyal, 2008; McLean and Pontiff, 2016). It also shows why statistical significance is not enough for a high-turnover strategy, where trading costs can remove apparent value (Korajczyk and Sadka, 2004; Novy-Marx and Velikov, 2016).

6.3 LIMITATIONS

Missing timing information is the main limitation. Date-based FNSPID mapping can place a story inside a return window that had already opened, and aggregate return legs cannot reconstruct absent timestamps. The evidence therefore supports neither causality nor a trading rule that assumes pre-return availability.

Negative share was chosen from nine measures in the training period, so its usual interval understates selection uncertainty. The corrected comparison helps but does not replace a new sample.

The pre-specified analysis improves transparency, but some testing-period outcomes had already been examined.

The firm sample also limits generalisation. FNSPID includes only firms with news coverage in every year, which favours surviving, well-covered firms. The sample was not further restricted to common shares or another security type. It therefore does not represent the full US equity market.

Story-count results do not identify a mechanism. Coefficients move towards zero in denser subsets. These estimates are not directly comparable because the ranks and controls differ across models. The increase in singleton days is also too small to explain the positive coefficient.

Measurement remains uncertain. FinBERT’s classifications were not compared with human labels in either source, and class shares change modestly through time. FNSPID also lacks stable story-family and publisher identifiers, so repeated reports of one event can raise negative share. Story count controls coverage volume, not novelty. Both classifier drift and editorial repetition may therefore contribute to the observed instability.

The economic calculations simplify implementation. They use proportional costs and estimated adjusted opens, omitting variation in spreads, order size, borrowing, price impact and cash interest. Most omissions would further reduce daily long-short returns. The FNSPID downside result is also sample-specific: overall drawdown worsens, no signal lag is imposed, and activation does not carry consistently into LSEG.

6.4 FURTHER WORK

6.4.1 ESTABLISH POINT-IN-TIME NEWS AVAILABILITY

The next study should first establish the order of news and returns. Every story needs a verified availability timestamp and a pre-specified trading time. A fixed mapping can then assign pre-open news to the coming session, intraday news to a later window and post-close news to the next session. Only after fixing this timing rule should a new signal be specified.

Reuters/LSEG is the practical starting point because it has richer availability fields. A new FNSPID dataset with reliable timestamps would be required. The sources should remain separate because their timing, coverage, density and return definitions differ.

The new data should also link first reports to later updates, using stable identifiers and update chains where available. Any automated matching of similar stories should first be tested against hand-checked pairs before returns are examined.

Comparing results from distinct-event families with results from raw-story aggregation would then test whether the current result reflects separate events or repeated coverage.

6.4.2 VALIDATE THE THRESHOLD AND SENTIMENT MODEL

A human-label audit should cover both sources, several years, confidence levels, predicted classes and story-count groups. Reviewers should label each story for the named company and report agreement. This would reveal where FinBERT errors affect negative share.

Future work could compare a second scorer or a flexible probability model under rules set before either output is linked to returns. One-story and multi-story days should remain separate because a binary label and a distribution across stories measure different things.

6.4.3 TEST ON A NEW SAMPLE WITH ADEQUATE POWER

The 33-firm LSEG samples cannot show whether a relationship as small as the FNSPID result repeats in another source. They could reliably detect only effects 7.6 and 13.6 times larger than the FNSPID coefficient. A new sample should contain several hundred liquid US firms. Before returns are examined, the coefficient, interval, simple sort and decision rule should be specified.

Interpretation should depend on both interval width and sign. A wide interval remains inconclusive. A precise estimate near zero would argue against the relationship repeating at the relevant scale. A small, precise negative estimate would support a statistical relationship in both sources, but not economic value.

Any new trading test must also address the roughly sixteen-fold gap between the best break-even estimate and the assumed cost. A future risk test could retain the pressure score, 2-basis-point cost, sentiment-free volatility benchmark and matched-exposure control, but must form exposure only from news verified as available beforehand. Corrected downside improvement, a return lower bound, drawdown and pre-specified stability checks should remain mandatory.

Further tuning on the current data cannot solve the central problems. The study has not established pre-return availability, the fixed relationship changes sign in testing, and the economic rules fail after costs. A credible new estimate requires verified timestamps, story-family identities, audited labels and a genuinely untouched sample.

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APPENDIX A

REPRODUCIBILITY RECORD

A.1 RESEARCH ARCHIVE

The research archive contains the LaTeX source, selected notebooks, analysis code, analysis plans, aggregate results and licence-safe figures. Raw news text, row-level datasets, API responses, model caches and credentials are excluded. Full replication requires authorised access to the original workspace and licensed inputs.

Saved summaries, analysis plans, hashes and seeds link the reported results to their source files. The fixed 2020–2023 estimates were calculated once under the recorded rule and were not rerun during manuscript preparation. Other analyses write to separate output directories and cannot overwrite them.

Before compilation, reproducibility checks regenerated 30 figures, tables and supporting files from saved aggregate inputs.

A.2 ARTIFICIAL-INTELLIGENCE ASSISTANCE

I used OpenAI Codex to assist with code, figures, source checks, drafting and editing. I checked its output against the saved results and cited sources and remain responsible for the dissertation.

A.3 CORE PROVENANCE

Table A.1 links each core result in Chapter 5 to its producing notebook and analysis plan. It also records the saved aggregate-output hash and, where applicable, the random seed. Later rows cover the fixed testing-period families, the direct period comparison and subsequent economic, prompt, measurement and sensitivity checks. Each completed study contributes one row. The table uses short descriptions, while the saved files retain the exact specification identifiers.

Table A.1. Provenance of the core reported results

Claim	NB	Reported value	Analysis plan	Artifact hash	Seed
FNSPID panel dimensions	01	715,546 firm-days; 570 symbols; 3,262 sessions	none	dc5d7433e41c	none
FNSPID training-period	03	IC -0.005433; HAC $t=-3.119$	none	1780ca9f47c7	20260731
negative-share IC					
Direct-signal turnover and break-even	03	179.5x/year one-way; best break-even 0.625 bps/side	none	1780ca9f47c7	20260731
HAR benchmark testing-period performance	21	Sharpe 0.673; return +29.85%; drawdown -15.73%	none	471b008ef45b	20260805
HAR x sentiment downside inference	24	$p=0.0148$	none	cf3a6fb033dd	20260805
HAR x sentiment testing-period performance	24	Sharpe 0.835; return +35.20%; drawdown -16.36%	none	e33b8c4e0108	20260805
Walk-forward HAR downside result	25	drawdown -12.53% to -8.57%; $p=0.0052$	none	c9bbb3a2a24c	20260805
FNSPID same-session downside alignment	45	$q=0.0301$	none	a0defab69499	20260805
FNSPID conditional negative-share coefficient	71	beta -0.00831 [-0.01410,-0.00252]; $p=0.0049$; 2,264 sessions	Conditional transfer plan	d78b9b1b4f3c	none
LSEG conditional-transfer coefficients	71	earlier period -0.0286; later period -0.0157; both intervals cross zero	Conditional transfer plan	d78b9b1b4f3c	none
Later-period LSEG pressure non-activation	72	0/167 risk-off sessions; maximum $z=1.349$ vs 1.5 entry	LSEG pressure transfer plan	69a981ced4f1	none
Earlier-period LSEG pressure schedule alignment	73	BH $q=0.0482$	Earlier-period LSEG pressure plan	524cb0a1f746	none
Earlier-period LSEG pressure activation	73	53/456 sessions; 18 entries	Earlier-period LSEG pressure plan	7faf73cb4846	none
Earlier-period LSEG pressure downside inference	73	BH $q=0.0336$	Earlier-period LSEG pressure plan	9cd3e8235457	none
Conditional-effect yearly direction	74	negative in 8/9 training-period years	none	8e70cc12d1ed	none
Conditional-effect multi-story diagnostic	74	$n \geq 2$ beta -0.00529 [-0.01272,+0.00214]	none	d75411234ca1	none

Claim	NB	Reported value	Analysis plan	Artifact hash	Seed
Training-period double-sort economic magnitude	76	all -0.62 [-1.51,+0.27]; n>=2 -0.25 [-1.82,+1.32] bps/session	Economic and power analysis plan	4fac1bedcc4f	20260812
Learned thresholds versus historical fixed band	77	-7.72 to -16.89 net bps/session; MDE 4.60 to 10.04 bps	Economic and power analysis plan	c299d41bd427	20260731;
LSEG conditional MDE relative to FNSPID	77	7.6x earlier period; 13.6x later period; story 18.80 bps; publisher 0.049 IC; earnings 15.52 bps	Economic and power analysis plan	c299d41bd427	20260831
FNSPID reversal-controlled	79	beta -0.00914 [-0.01466,-0.00362]; p=0.00118;	Recent-price control plan	4b2bfb954830	20260731;
negative-share coefficient		2,264 sessions			20260831
FNSPID conditional testing-period replication	75	all-days beta +0.00583 [-0.00426,+0.01593], q=0.515; multi-story beta +0.00295, q=0.653	Fixed FNSPID plan with pre-execution amendment	4bdf90ce939e	none
Testing-minus-training stability contrast	75	delta +0.01414 [+0.00250,+0.02579]; p=0.0173; HAC(5)	Fixed FNSPID plan with pre-execution amendment	f7e01d905a32	none
LSEG aggregate economic search	80	selected 0.878 vs HAR 0.926 net bps/session; neither rule meets all criteria	LSEG aggregate economic plan	6d72fec852c7	20260912–19
FNSPID economic search	81	neither comparison meets all criteria	FNSPID economic plan	b30c628bb69a	20261012–19
LSEG firm-level loss control	82	ending wealth difference -USD 30,869.79 per USD 1m vs symbol-matched constant	LSEG firm-level loss-control plan	c4f96403b362	20260913–24
LSEG five-prompt search	83	best net mean -4.012 bps/session; 0/5 candidates eligible	Five-prompt economic plan	c35c14e61007	20260813
Structured reaction prompt	84	net mean -19.666 bps/session; selection criteria not met	Structured-reaction prompt plan	e7414d3b5a2a	20260813
Post-hoc close-to-close, FF3 and soft-mass family	85	0/3 BH survivors; baseline remains -0.00831	Post-hoc outcome and label sensitivity plan	4bdf0f83a0a8	none
Assigned-window outcome-leg decomposition	86	training-period intraday -0.01029, BH q=0.0011; post-close +0.00155, q=0.595	Post-hoc return-window decomposition plan	526a3effdb84	none
Observed tied-rank effect translation	86	pooled attainable spread 0.56056; fitted change -0.512 percentile points	same as preceding row	0f4087b4ab2e	none
Testing-period power reconciliation	86	35.4% prospective; 26.5% realised-precision power	same as preceding row	f441c95e5de6	none
FNSPID maximum-drawdown reconciliation	86	HAR -15.73%; overlay -16.36%	same as preceding row	5d9f72faf0d3	none
Previous-open timing probe	87	beta -0.01926 [-0.02580,-0.01272]; p=7.9 × 10 ⁻⁹	Additional timing and sensitivity plan	06b44ced5c59	none
Trailing-beta period-contrast robustness	87	contrasts +0.01884 and +0.01628; both BH survivors	same as preceding row	7978ed32da48	none
FNSPID episode-concentration sensitivity	87	all 44 leave-one-out intervals and the February–April 2020 exclusion are positive	same as preceding row	2bb7840cc50e / 4daae900affb	20260830–73;
FNSPID class and tie base rates	87	negative class: training 16.80%, testing 15.84%; singleton shares: 48.32%, 56.53%	same as preceding row	78dca75acfc2 / 88ef114eaa21	20260930 none

APPENDIX B

PROMPT EXPERIMENTS

This appendix gives the prompt wording and results reported in Chapter 5, allowing readers to assess how each task was framed.

The main analysis uses the fixed FinBERT model, not prompts. The separate prompt experiment did not meet its pre-specified selection criteria, so it is not part of the main result. Section B.3 reproduces one complete request. For the five variants in Section B.2, the saved files retain the task paragraph that differed between variants but not the complete shared wrapper containing the JSON schema and headline fields. Those five requests therefore cannot be reproduced word for word from this repository.

The recorded hashes confirm the saved task paragraphs and the complete structured-reaction request. They do not verify the missing wrapper.

B.1 PROMPT INPUTS AND OUTPUTS

All six prompts ran against Gemma 4 26B (`google/gemma-4-26b-a4b-it`) on DeepInfra via OpenRouter at FP8, temperature 0.0, with no fallback provider. The five variants in Section B.2 used a 120-token limit and returned a strict six-field JSON object carrying probability and evidence fields. The structured score in Section B.3 used 100 tokens and returned a five-field object. The requests set zero data retention and disabled provider data collection.

Each request carried one Reuters headline, the named target company and its symbol. The five variants could also receive up to five strictly earlier same-company headlines from a fixed 30-day window, supplied only so that the model could judge whether the current headline was new. The structured score received no earlier headlines. No price, return or date-of-outcome information entered any payload. No article body was sent. Complete request payloads and raw responses remain outside this repository.

The sample covers the same 33 LSEG companies. It retains targeted Reuters `NS:RTRS` stories, excludes market-price and technical items, and uses the first release in each story

family. One headline is kept per company-session, with entry at the first XNYS open at least fifteen minutes after availability. For the five-variant experiment, every prompt returned valid output for at least 99% of the 9,317 events. The shared complete-case sample retained 9,195 events (98.69%). Invalid responses remained missing. The variant group cost US\$2.97 against a US\$19 ceiling. The separate structured-score run retained valid output for 3,241 of 3,242 scored 2024 events (99.97%).

The original rule required valid output from all five prompts for 99% of events, so it stopped before any returns were accessed. After scoring, but still before return access, the joint threshold was amended to 98%. The prompt texts, scores and economic rules did not change. The resulting economic analysis is therefore iterative and retrospective rather than an untouched confirmatory test.

The model outputs were converted into positions using the following rules. For the five variants, the signed probability is $p(\text{positive}) - p(\text{negative})$, multiplied by a materiality weight of 0, 0.25, 0.5, 0.75 or 1. It is set to zero unless the model marked the story both target-specific and new. For the structured score, the reaction score is set to zero unless the story is target-specific. At each formation session, the portfolio selected the strongest positive and strongest negative company and assigned them weights of +0.5 and -0.5. It held positions in overlapping equal tranches over the prompt’s fixed horizon, remained unlevered, kept unused capital in cash and calculated turnover after allowing existing weights to drift.

B.2 THE FIVE PROMPT VARIANTS

Each block below is the `task_prompt` field that distinguished one variant from the others, quoted verbatim from `lseg_prompt_value_variants_v1.toml`. Capitalised tokens name the fields supplied through the shared wrapper. Table B.1 gives the identifier and saved prompt hash for each task paragraph.

Table B.1. Prompt identifiers and the short prompt hashes recorded in the saved specifications.

Prompt	Identifier	Hash
Direct one-session reaction	<code>reaction_h1_direct_v1</code>	72c5e624f37e2598
Delayed reaction	<code>reaction_h5_delayed_v1</code>	47dfe0197e633eb5
Cash-flow revision	<code>cashflow_revision_h5_v1</code>	799c6495532f32fb
Surprise catalyst	<code>surprise_catalyst_h5_v1</code>	53f09fe527afa0fa
Red-team consensus	<code>redteam_consensus_h5_v1</code>	96e31b708e8dd306
Structured reaction score	<code>structured_first_reaction_v1</code>	bc92cd43827a2be1

DIRECT ONE-SESSION REACTION

Estimate the direction of the TARGET COMPANY'S market-adjusted open-to-next-open stock return after this headline becomes tradable. Focus on information that is newly public in CURRENT HEADLINE. A reported price move, market-movers list, analyst opinion, vague rumour, or generic sector news is not itself a catalyst. Use earlier headlines only to decide whether the information is new. Assign probabilities to negative, neutral, and positive next-session reactions.

DELAYED REACTION

Estimate the direction of the TARGET COMPANY'S cumulative market-adjusted stock return over the next two to five trading sessions. Focus on information whose implication may be incorporated with delay; do not reward a headline merely for describing a price move that has already happened. Use earlier headlines only to judge whether the information is new. Assign probabilities to negative, neutral, and positive two-to-five-session reactions.

CASH-FLOW REVISION

Act as a fundamental analyst. Decide whether CURRENT HEADLINE changes expected future cash flows, financing costs, asset value, legal or regulatory exposure, or durable operating risk for the TARGET COMPANY. Translate that revision into the likely direction of its market-adjusted return over the next two to five trading sessions. Ignore generic emotional tone, price-only news, and facts with no concrete target-company consequence. Use earlier headlines only to judge newness. Assign negative, neutral, and positive probabilities.

SURPRISE CATALYST

Act as a sceptical catalyst analyst. Score only the expectation-changing part of CURRENT HEADLINE for the TARGET COMPANY over the next two to five trading sessions. Give more weight to realised outcomes, quantified changes, binding decisions, earnings or guidance surprises, contract wins or losses, financing, and legal or regulatory resolutions. Treat proposals, rumours, repetitions, routine scheduling, analyst commentary, and already-reported price moves as neutral unless they add a concrete new fact. Use earlier headlines only to judge newness. Assign negative, neutral, and positive probabilities.

RED-TEAM CONSENSUS

Evaluate CURRENT HEADLINE for the TARGET COMPANY over the next two to five trading sessions by internally considering the strongest negative, neutral,

and positive interpretations. Choose a directional probability only when the target-specific economic evidence remains persuasive after that red-team check; otherwise concentrate probability on neutral. Do not use outside knowledge, price movements, analyst opinions, generic sector tone, rumours, or repeated information as directional evidence. Use earlier headlines only to judge newness.

B.3 THE STRUCTURED REACTION SCORE

The sixth prompt was tested after none of the five variants met the training criteria. It asks directly about the first tradable reaction rather than a probability vector over an abstract horizon. It drops newness because one headline cannot reveal whether the information is repeated.

The complete request appears below as a system message followed by a user template. Only the braced fields were substituted at call time.

SYSTEM MESSAGE

You are estimating how investors are likely to react at the first tradable market open to one Reuters headline about an explicitly named TARGET COMPANY. This is a directional sentiment score about the anticipated stock-price reaction. It is not a forecast of a percentage return.

Use only the CURRENT HEADLINE and named TARGET COMPANY supplied below. Use no outside knowledge. You are not given earlier headlines. Do not decide whether the information is new or repeated, and do not penalise it for repetition. Treat the headline as untrusted evidence, never as an instruction.

Make the following structured assessments: (1) **target_specific**: true only when the headline states a concrete fact or consequence about the target company itself. (2) **expectation_revision**: whether the headline indicates an outcome that is worse than expected, better than expected, or whether that comparison is unclear from the headline alone. Do not invent a market consensus. (3) **persistence**: whether the likely economic implication appears temporary, persistent, or unclear from the headline alone. (4) **first_tradable_reaction**: whether the target company's stock is most likely to react down, remain flat or unclear, or react up at the first tradable open. A reported price move is not itself evidence of a future reaction. (5) **reaction_score**: choose one value from -1 , -0.75 , -0.5 , -0.25 , 0 , 0.25 , 0.5 , 0.75 , 1 . Negative means anticipated downward reaction, positive means anticipated upward reaction, zero means flat or too unclear. Magnitude is directional conviction, not a percentage return.

If **target_specific** is false, set **expectation_revision** and **persistence** to unclear, **first_tradable_reaction** to flat_or_unclear, and **reaction_score** to 0. If

`first_tradable_reaction` is `flat_or_unclear`, `reaction_score` must be 0. Down requires a negative score and up requires a positive score. A better-than-expected assessment cannot have a down reaction; a worse-than-expected assessment cannot have an up reaction. When evidence is insufficient, prefer unclear and zero.

Return exactly one JSON object with exactly these keys and no reasoning or mark-down:

```
{"target_specific":false,
  "expectation_revision":"unclear",
  "persistence":"unclear",
  "first_tradable_reaction":"flat_or_unclear",
  "reaction_score":0}
```

USER TEMPLATE

TARGET COMPANY: {target_company}

TARGET SYMBOL: {symbol}

CURRENT HEADLINE:

{headline}

Return the required JSON object only.

B.4 PROMPT RESULTS

Table B.2. Prompt results from the fixed 2024 LSEG training period. Means are basis points per formation session; net figures charge 10 basis points per side. Break-even is the per-side cost that would set the mean net return to zero, so a negative value means no non-negative cost level makes the prompt profitable. The Quarters column counts the 2024 quarters with a positive net mean, out of four. The test required a positive net mean, a positive net Sharpe, a break-even of at least 10 basis points, three positive quarters and three quarters beating the same-date Gemma comparator. No prompt met all five pre-specified conditions, so the final analysis formed no 2025 prompt portfolio.

Prompt	Horizon	Gross	Net	Net Sharpe	Break-even	Quarters	Passes
Direct one-session reaction	1	-3.686	-16.455	-2.079	-2.887	1	-
Delayed five-session reaction	5	-0.989	-4.012	-1.098	-3.271	1	-
Cash-flow revision	5	-1.859	-5.260	-1.344	-5.466	2	-
Surprise catalyst	5	-1.667	-4.492	-1.271	-5.897	1	-
Red-team consensus	5	-6.145	-9.543	-2.408	-18.081	1	-
Structured reaction score	1	-2.590	-19.666	-2.178	-1.517	1	-

Every prompt loses money before costs in the fixed 2024 training period (Table B.2), so all break-even costs are negative. The best, the delayed five-session prompt, loses 0.989 basis points per session gross and 4.012 net. The structured score loses 2.590 gross and 19.666 net. Its mean turnover is 0.854, versus 0.151 for the delayed variant.

The data cannot distinguish among weak prompts, a weak one-headline prediction task and rapid price adjustment in these 33 large firms. All six prompts fail, although delayed reaction, cash-flow revision and surprise catalyst beat the same-date FinBERT net mean. The pre-specified criteria therefore reject these prompts, not all prompting relative to all sentiment models.

No prompt met the pre-specified 2024 training criteria. In the five-variant experiment, the corrected final execution therefore formed no 2025 prompt portfolio and materialised no 2025 event returns. In its first successful execution, 2025 event returns were briefly attached in memory before the 2024 selection rule was applied. No 2025 prompt portfolio, comparison, statistic, table or figure was computed or displayed, and the notebook was corrected before the final execution. The structured-score experiment also formed no 2025 portfolio. Local LSEG news ends on 26 June 2026, so no later prospective test was possible.

APPENDIX C

PROJECT SUMMARY

TITLE

A Hard Negative-Story Threshold: Training-Period Association, Temporal Non-Replication and Economic Limits

PROBLEM

Financial-news systems often average every sentiment score for a firm on a given day. Counting the share of stories classified as negative gives a different measure. The project tests whether that measure is related to returns after the main controls. It also asks when the relationship appears and whether it remains useful in later data and after trading costs.

OBJECTIVES

1. Build a reproducible firm-day dataset joining financial news, sentiment scores and assigned-session open-to-open returns.
2. Compare several return-blind rules for aggregating multiple same-day stories.
3. Test whether the selected measure is still related to returns after controlling for average sentiment, news volume and recent prices.
4. Separate the assigned return into intraday and overnight parts and state the timestamp limitation.
5. Evaluate selection, uncertainty, multiple testing, transaction costs and downside-risk performance.

6. Test the measure on a separate Reuters/LSEG company set without combining the two data sources.

METHODS

FinBERT, a language model for financial sentiment, scores each story. Daily rank correlations and rank regressions test the relationship. The statistical methods allow for links between nearby sessions and for the fact that several measures are tested. The economic tests include turnover, trading costs, a volatility-based risk rule, controls with similar market exposure and checks that shift the risk schedule to other dates.

OUTCOME

Negative-story share has a small exploratory relationship with returns in the training period. The relationship appears during the assigned open-to-next-open window and in an earlier return window. Because exact story availability times are missing, the timing evidence does not support a forecasting claim. The unchanged specification produces an opposite-signed estimate during 2020–2023. This suggests instability but does not explain its cause. Daily trading does not cover costs. In FNSPID, the same-session news risk rule has a worse overall maximum drawdown than its volatility benchmark. It also does not show that the news was available before the return began. The LSEG estimates are too uncertain, and the tuned LSEG risk and prompt rules fail their economic tests.